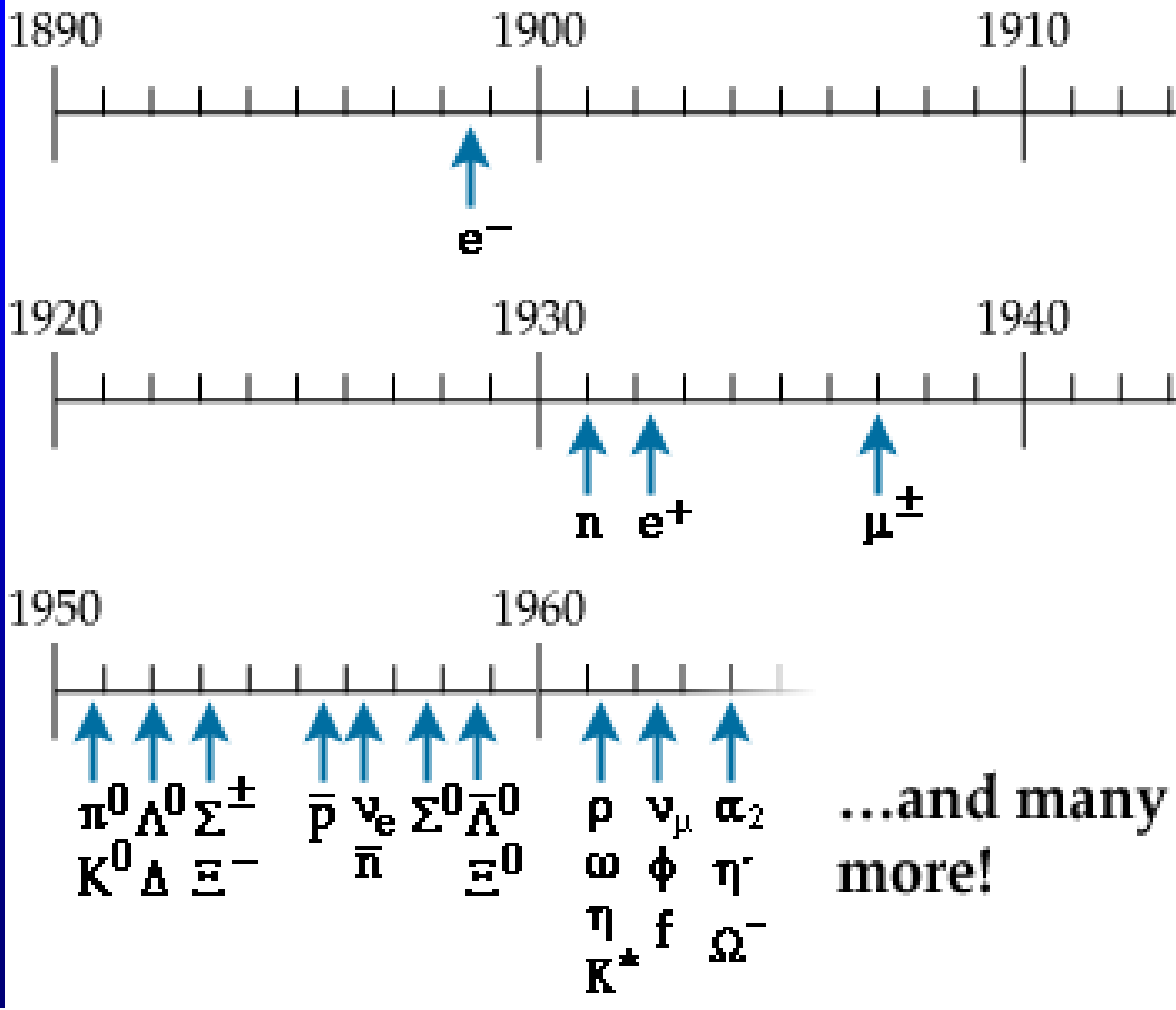
Chapter 30 Leptons and Quarks

30.1 Particles and their classification

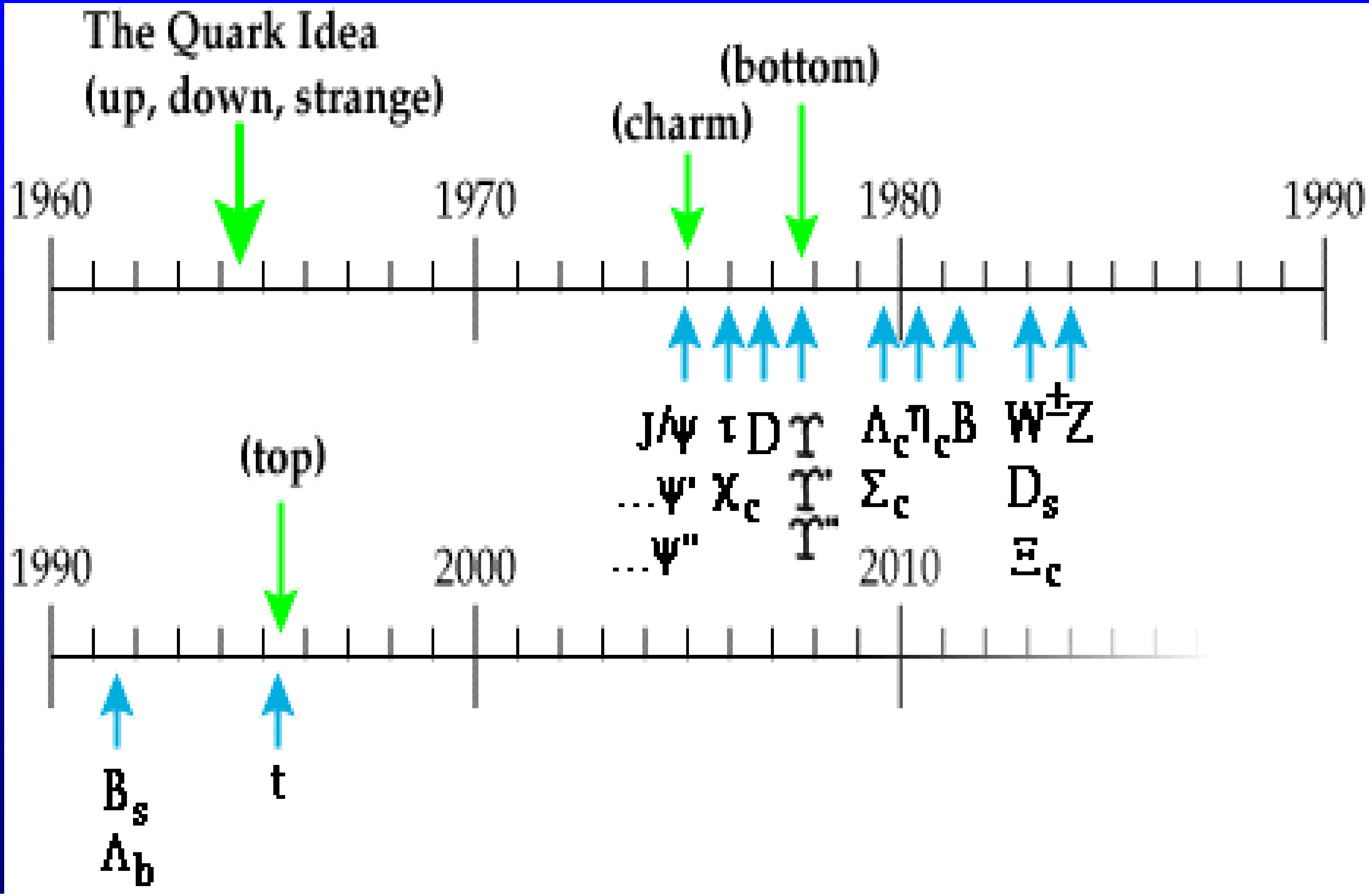
Murray Gell-Mann was awarded 1969 Nobel Prize for his contributions and discoveries concerning the classification of the elementary particles and their interactions.

Zoo of particles

Particles discovered 1898 - 1964



Zoo of particles   Particles discovered 1964-



Classified mainly according to mass, spin, interaction and lifetime criteria have partly been incorporated

	particle	rest mass/MeV	spin	strangeness
Photon	γ	0	1	0
Lepton	e^-	0.511	**1/2**	0
	$\boxed{\mu^-}$	105.7	**1/2**	0
	τ^-	1777	**1/2**	0
	ν	$\sim 10^{-6}$	**1/2**	0
meson	$\pi^\pm$	136.9	0	0
	π^0	135.0	0	0
	$K^\pm$	493.7	0	± 1

	K	497.7	0	+1
	J/Ψ	3.1×10^3	1	0
nucleon	p	938.3	1/2	0
	n	939.6	1/2	0
	$\boxed{\Delta^+}$	1232	3/2	0
hyperon	Λ^0	1115.6	1/2	–1
	Σ^+	1189.4	1/2	–1
	Ω^-	1672.2	3/2	–3

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	K^0	497.7	0	+1
	J/ψ	3.1×10^3	1	0
nucleon	p	938.3	1/2	0
	n	939.6	1/2	0
	Δ^+	1232	3/2	0
hyperon	Λ^0	1 115.6	1/2	–1
	Σ^+	1 189.4	1/2	–1
	Ω^-	1 672.2	3/2	–3

nucleons are not simple elementary point particles

(1) $\mu_p \approx 2.78\ \mu_N$, $\mu_n \approx -1.93\ \mu_N$

(2) Nuclear force ~ van der Waals' force

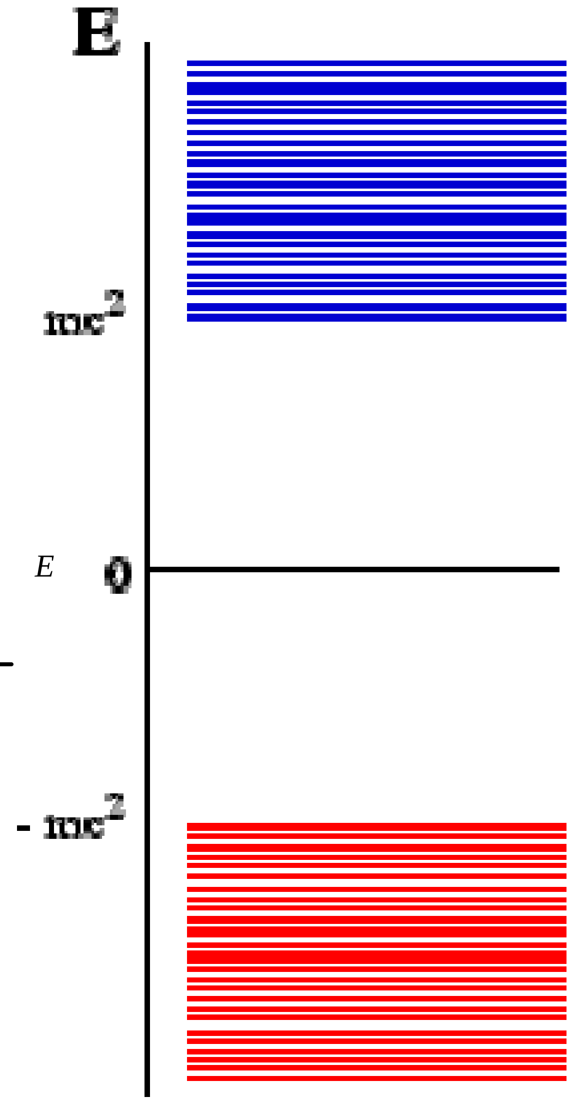
(3) Nucleon's excited state: Δ etc. discovered.

(4) Elastic scattering of electrons by nucleon indicates that nucleon's charge is of a finite radius of about 0.8 fm. (R. Hofstadter 1961 Nobel Prize)

(5) In deep inelastic scattering—"nuclear Rutherford scattering", $\leq 10^{-18}$m
SLAC 20GeV Friedman , Kendall, Taylor
1990 noble prize

From the Dirac equations:

$$E = \pm\sqrt{m^2c^4 + p^2c^2}$$



Elementary particles are also classified as particles and antiparticles.

Electron-positron (e^+)

Proton-antiproton ($\bar{p}$)

Neutron-antineutron

Photon-photon

From decay modes:

$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu$$ particle

$$\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$$

$$\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$$

$$\pi^+ \rightarrow \mu^+ + \nu_\mu$$

π^+, π^0, π^- particles set

π^-, π^0, π^+ antiparticles set classification

(1) leptons s = 1/2.

(2) quarks s = 1/2. “stratum”

qqq = baryon (Fermion)	p, n, Δ, ⋯
$q\bar{q}$ = meson(Boson)	π, K, D, F

(3) field quanta (field particles) s = 1 or s = 2

photon

gluons

intermediate bosons $W^{\pm}, Z^{0}; m \neq 0$

graviton

- leptons
  - e family
  - μ family
  - τ family
- quarks
  - mesons
    - K
    - π, ρ, w, J/ψ
  - baryons
    - nucleons p, n, D
    - hyperons
- field quanta
  - photons
  - gluons
  - $W^{\pm}$, Z^0
  - gravitons

hadrons

leptons

electron   muon   tau lepton   neutrinos

hadrons

baryons (fermions)   mesons(bosons)

nucleons   hyperons(strange)   heavy mesons (strange)

particles that mediate forces (field particles)

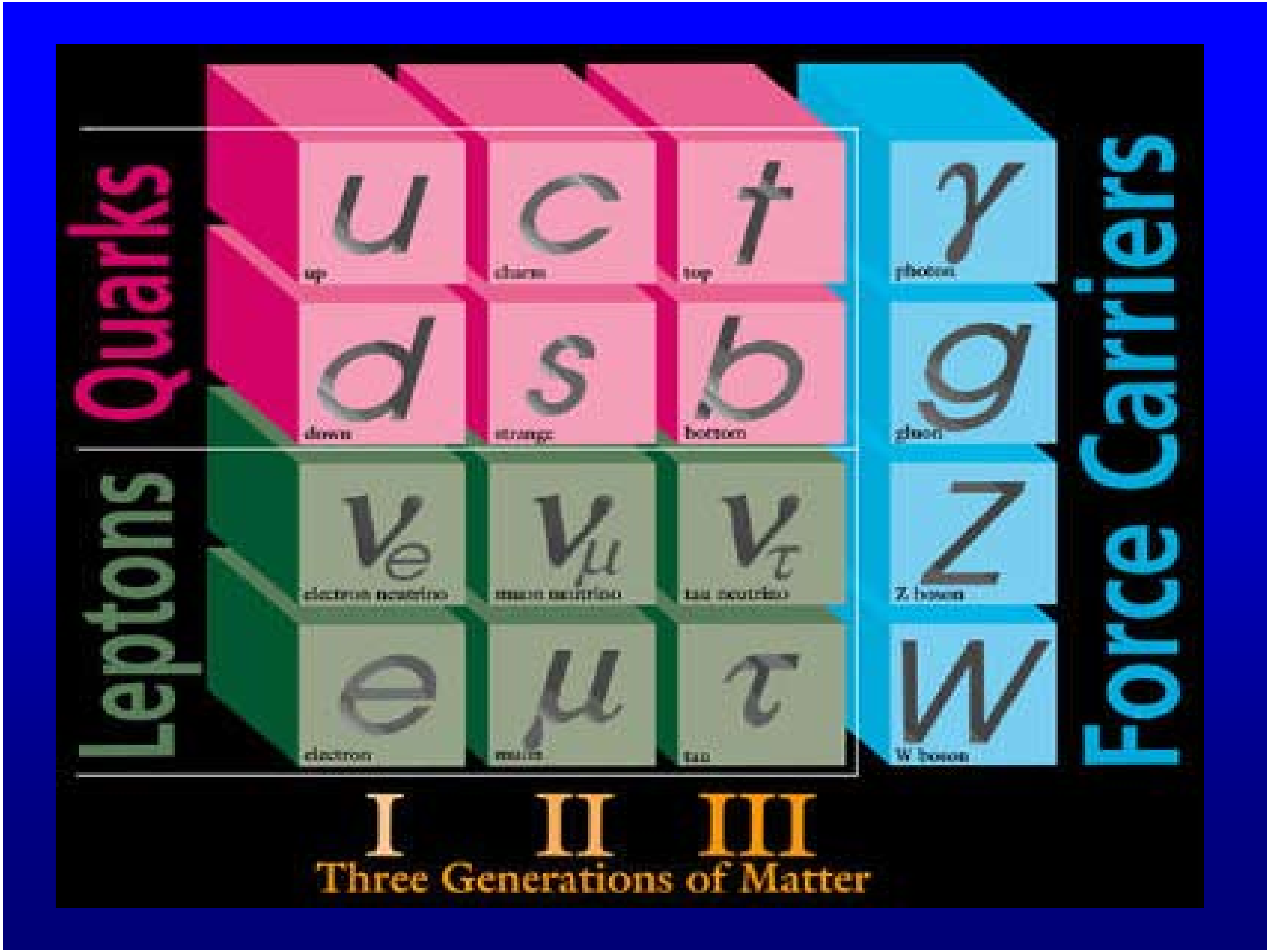
photon,   intermediate vector bosons,   gluons, graviton

In 1964, Gell-Mann and George Zweig proposed quark model, first three favors u, d, s appeared. In 1974 the discovery of J/ψ led to charmed quark c.

At that time known leptons were e , ν_e , μ , ν_μ

In 1975 not so light lepton τ was found

In 1977 and 1994 two more quarks were proposed: bottom or beauty quark b and top or truth quark t.


Quarks
Leptons
Force Carriers
u
up
c
charm
t
top
γ
photon
d
down
s
strange
b
bottom
g
gluon
ν_e
electron neutrino
ν_μ
muon neutrino
ν_τ
tau neutrino
Z
Z boson
e
electron
μ
muon
τ
tau
W
W boson
I
II
III
Three Generations of Matter

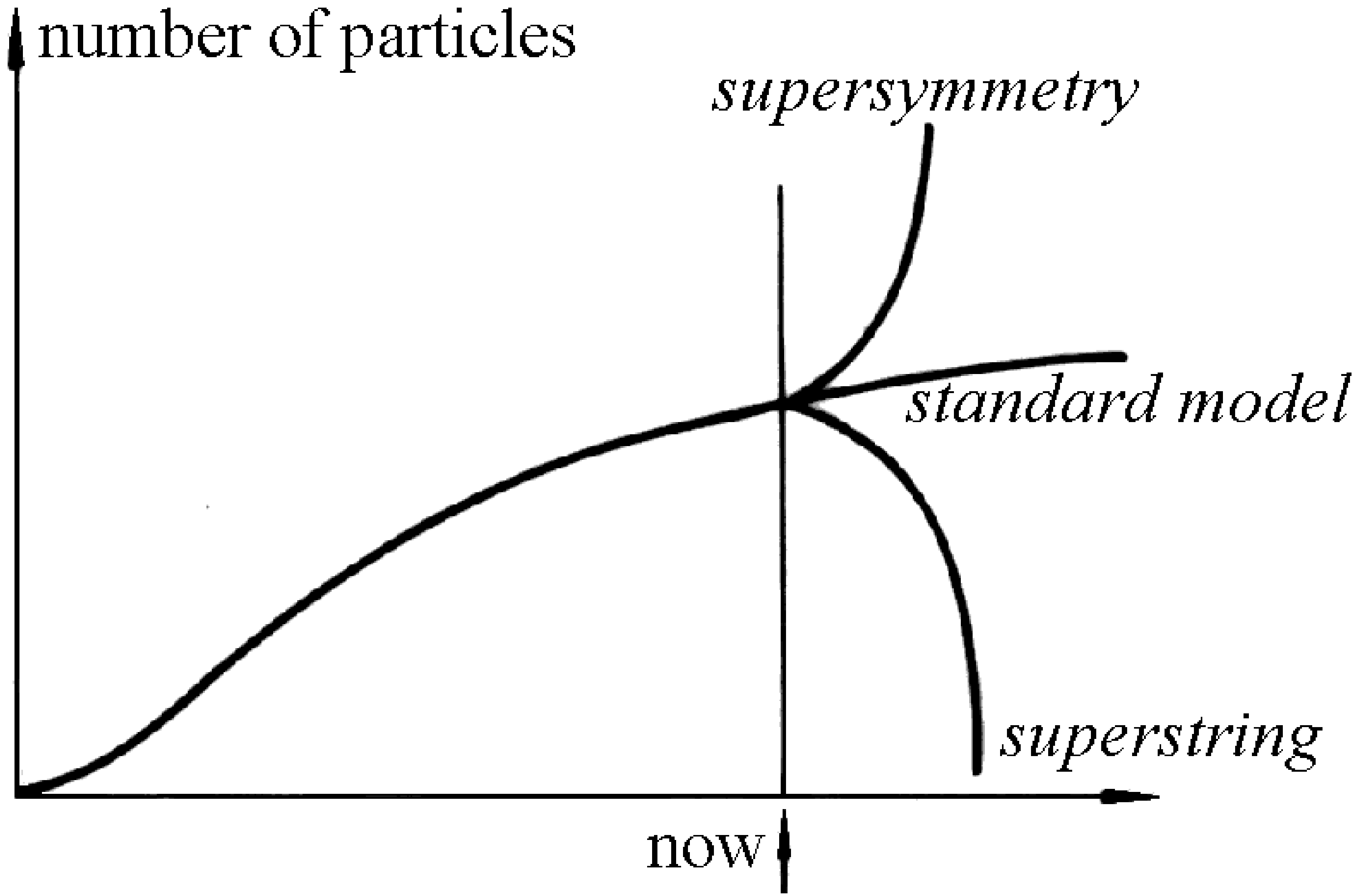
12 leptons,

36 quarks (color RGB and anticolor),

photon, intermediate bosons, and 8 gluons are 12.

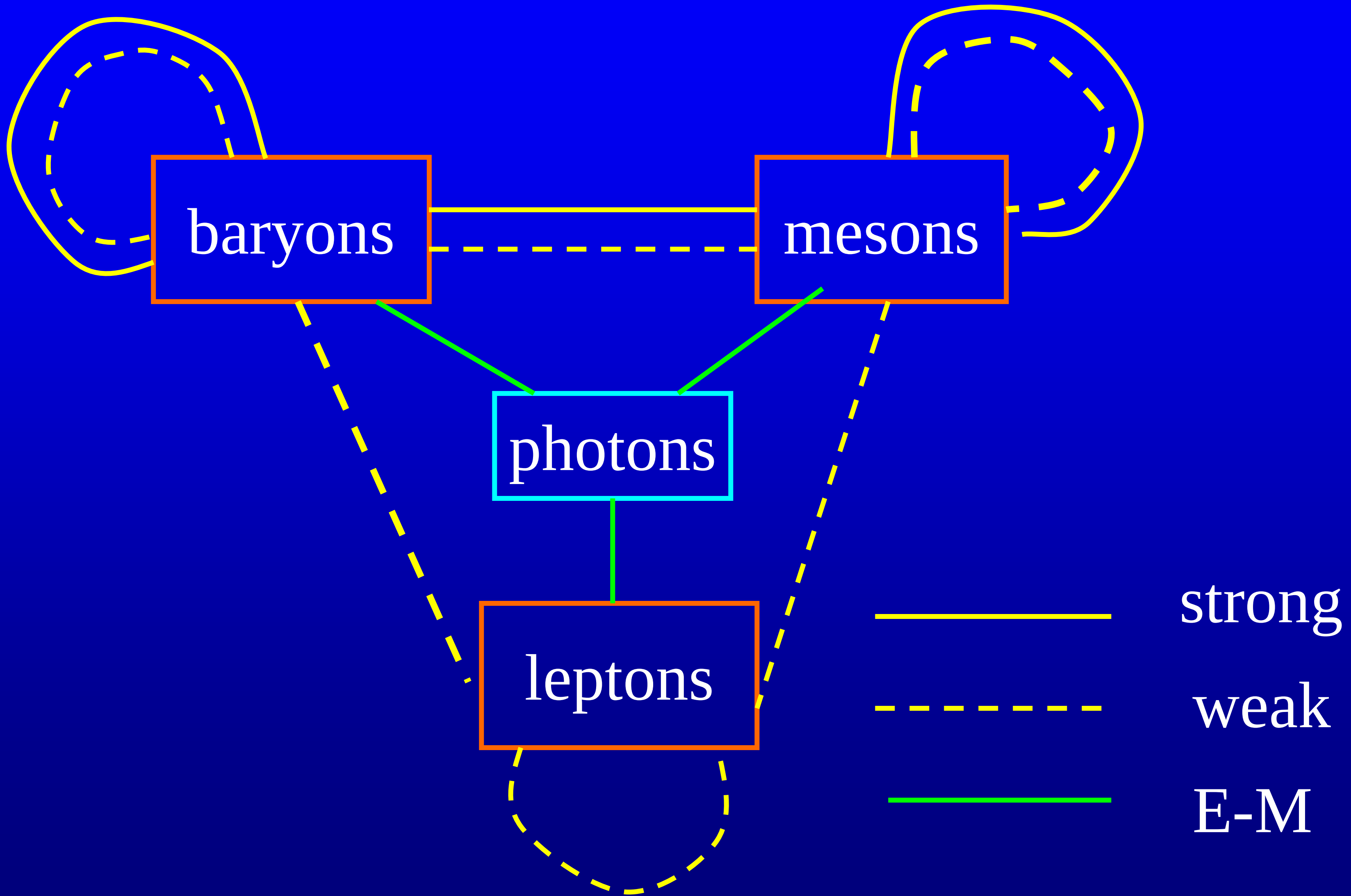
The total is 60.

Higgs particle and graviton.



Mr. Glashow’s prediction(1995,6,19)

30.2 Interactions



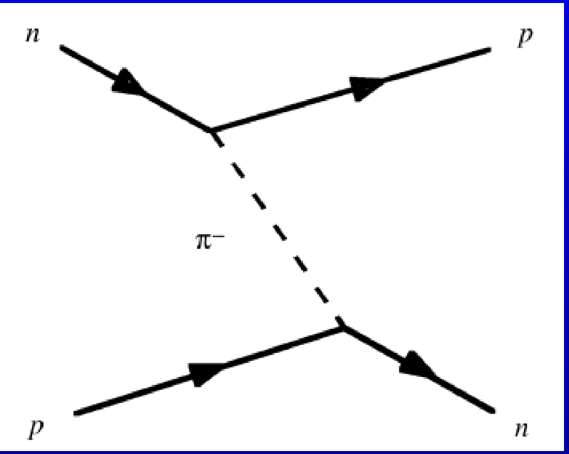
E-M interactions: Quantum electrodynamics (QED)

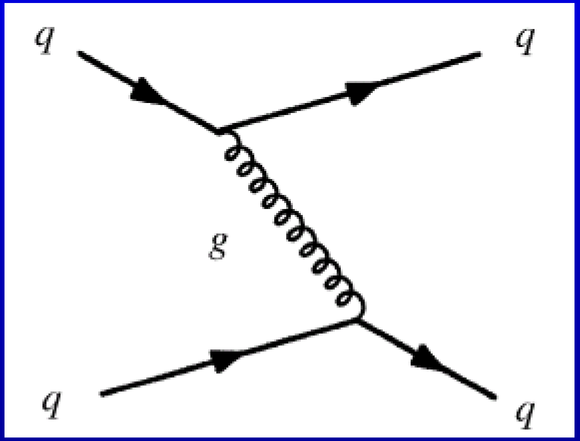
Weak interactions: Gauge Field Theory
GWS model of electroweak interactions

Strong interactions : Quantum chromodynamics (QCD)

Strong interactions :
the interaction between nucleons, nucleons and mesons and a number of other particles.

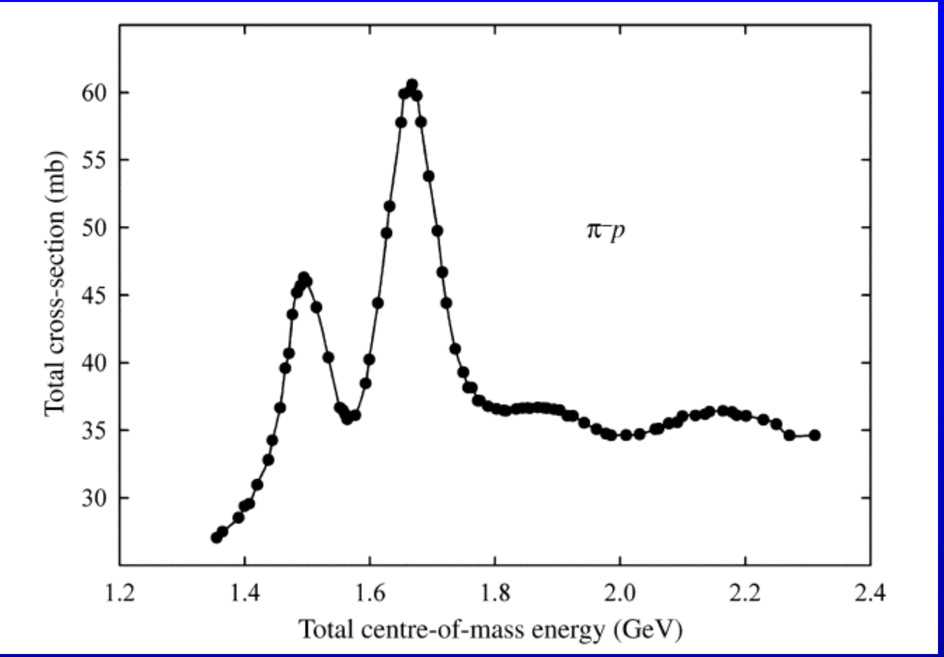
On the nuclear level the mesons act as the quanta of the strong interaction.





On the subnuclear scale the interactions are mediated by the exchange of gluons between quarks.

the energy level width of the resonance Δ is about $\Delta E \approx 300\text{MeV}$

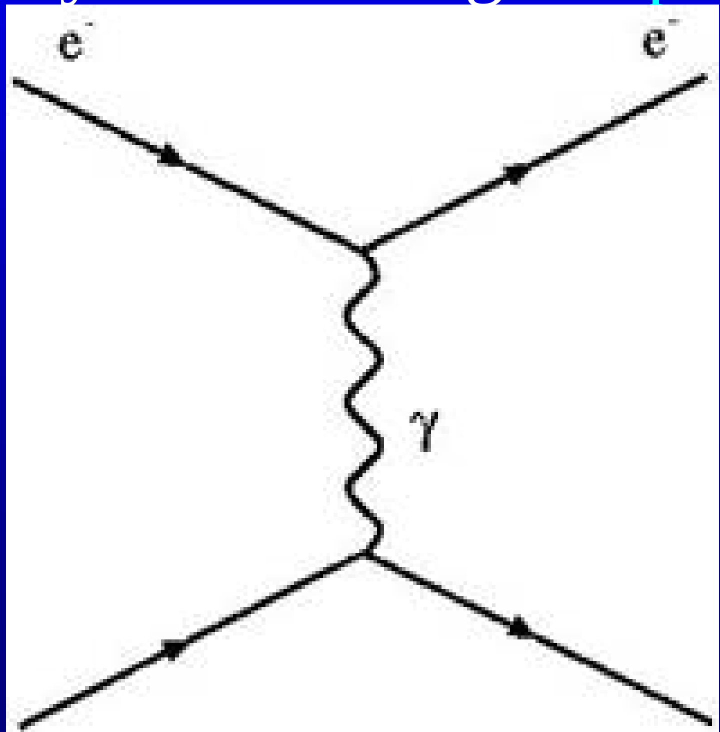


The time scale of hadron dynamics is

$$\tau \sim \frac{\hbar}{\Delta E} \sim \frac{10^{-16}\text{eV}\cdot\text{s}}{300\text{MeV}} \sim 10^{-23}\text{s}$$

Electromagnetic interactions are responsible for force between electrically charged particles and

are mediated by the exchange of photons.



Weak process is responsible for many particle decays such as radioactive decays (the basic process)

$$n \rightarrow p + e^- + \bar{\nu}_e$$

pion and muon decay, and a number of other decay processes.

Beta decay is a process that is very slow compared to other nuclear process: while in the fastest instances the lifetime is of order milliseconds, in the slowest cases the lifetime is greater than the age of the universe.

The cross-section for reactions induced by the low energy neutrinos produced by reactors are exceedingly small: of order 10^{-18}fm^2, corresponding to a mean free path of order 10^{14}km in solid rock.

Gravitational interactions exist between all particles have mass and

are believed to be mediated by the so-far-undetected gravitons.

The strength

Strong : 1

Electromagnetic :

$$\frac{1}{4\pi\varepsilon_0}\frac{e^2}{\boxed{\hbar/mc}} : mc^2 = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{\hbar c} \equiv \alpha$$

Compton wavelength

$$\alpha = \frac{1}{137} \sim 10^{-2}$$

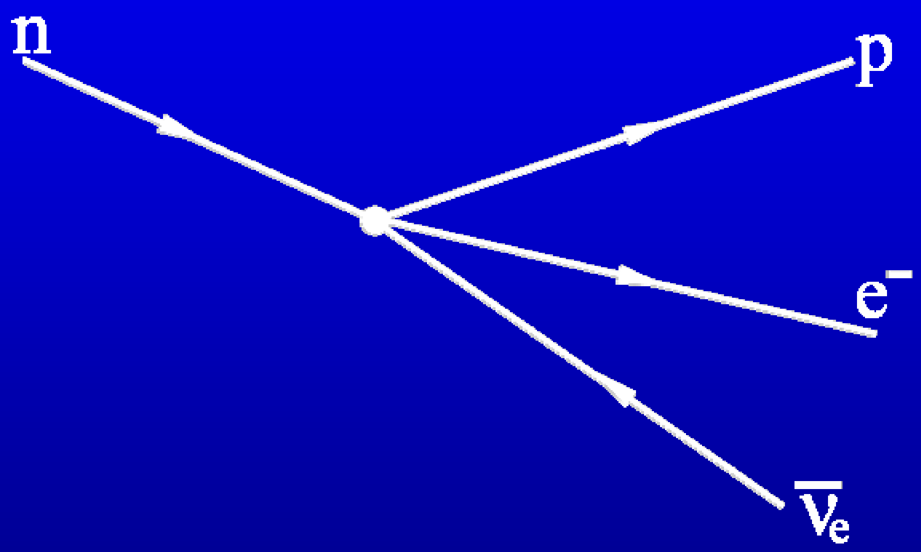
Weak : 10^{-5} or 10^{-10}

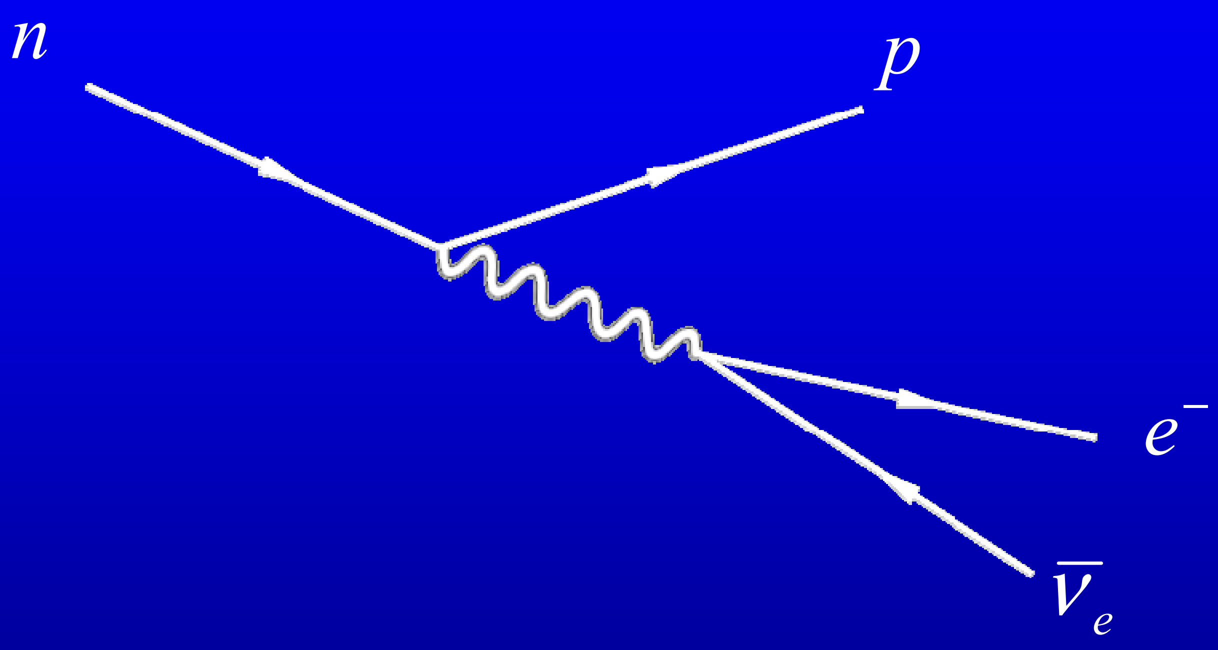
Gravitational :

$$\frac{Gm_{\mathrm{p}}^{2}}{\hbar/m_{\mathrm{p}}c} : m_{\mathrm{p}}c^{2} = \frac{Gm_{\mathrm{p}}^{2}}{\hbar c} \sim 10^{-39}$$

Electroweak

According to universal Fermi theory, weak interaction is a point action(four fermions interaction)





similar to electromagnetic interaction.
What are the field quanta?

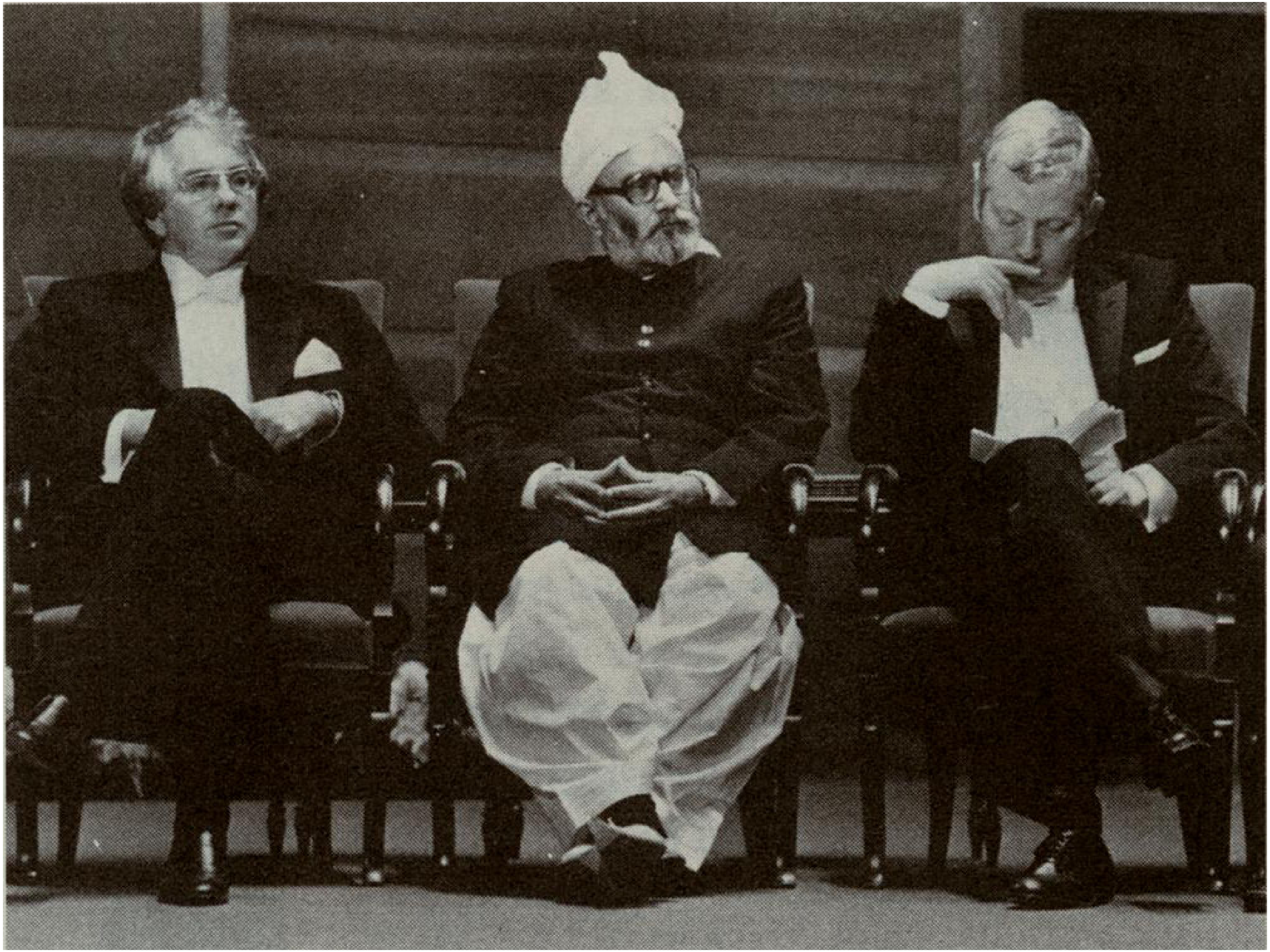
During 1961~8, S. Glashow (1961), S. Weinberg (1967) and A. Salam (1968) established "standard model" and united the electromagnetic and weak interactions.

intermediate vector bosons $W^{\pm}$

$$m_{W} \sim 83.0^{+3.0}_{-2.8}\ \mathrm{GeV}/c^{2}$$

Electroweak theory predicted a neutral current weak interaction propagated by Z^{0}

$$m_{Z^{0}} \sim 93.8^{+2.5}_{-2.4}\ \mathrm{GeV}/c^{2}$$



Groups UA1 and UA2 of CERN (the European Organization for Nuclear Research), reported their results on August 1983.

$$m_{\mathrm{W}} = (82.1 \pm 1.7)\,\mathrm{GeV}/c^2$$

$$m_{\mathrm{Z}^0} = (93.0 \pm 1.7)\,\mathrm{GeV}/c^2$$

The name CERN is derived from the acronym for the French Conseil Européen pour la Recherche Nucléaire, or European Council for Nuclear Research, a provisional body founded in 1952 with the mandate of establishing a world-class fundamental physics research organization in Europe.

S. L. Glashow, A. Salam and S. Weinberg won the 1979 Nobel prize
for their contributions to the theory of the unified weak and electromagnetic interaction between elementary particles, including inter alia the prediction of the weak neutral current

The 1984 Nobel prize in physics was awarded to Carlo Rubbia and Simon Van der Meer

for their decisive contributions to the largest project, which led to the discovery of the field particles $W^{\pm}$ and Z^0.

The strength of weak interaction will be similar to electromagnetic when

$$mc^2 > 10^2\,\mathrm{GeV}, \qquad d < 10^{-18}\,\mathrm{m}$$

30.3 Conservation laws

conservation laws of momentum,
angular momentum,
charge and
mass-energy.

There is no law of rest-mass conservation

(1) Conservation of lepton number L_e, L_μ, L_τ

The electron and electron neutrinos are assigned an electron lepton number $L_e = +1$,
while positron and $\bar{\nu}_e$ are assigned –1.

The muons and its corresponding neutrinos are assigned $L_\mu = +1$, and their antiparticles –1.

The tauons and tau(on) neutrinos have tau lepton number $L_\tau = +1$, and antiparticles –1.

$$n \rightarrow p + e^- + \bar{\nu}_e + 0.8\,\mathrm{MeV}$$

$$\mu^- \rightarrow \nu_\mu + e^- + \bar{\nu}_e$$

The following modes are forbidden:

$$e^- + p \rightarrow n + \bar{\nu}_e$$

$$p \rightarrow e^+ + \gamma$$

Example problems:

$$\mu^+ + e^- \rightarrow \bar{\nu}_\mu + \nu_e$$

$$\mu^+ \rightarrow e^+ + \nu_e + X \qquad \mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$$

(2) Conservation of baryon number *B*

Nucleons and hyperons are assigned $B = +1$, and anti-baryons -1

$$p\ p \rightarrow p\ p\ p\ \overline{p}$$

$$p\ p \rightarrow \begin{cases} p\ n\ \pi^{+} \\ p\ p\ \pi^{0} \\ p\ n\ \pi^{+}\ \pi^{0} \\ p\ p\ \pi^{+}\ \pi^{-} \end{cases} \quad \text{Ok}$$

$p\,p \rightarrow p\,\pi^+$

$p\,p \rightarrow p\,p\,n$

violate the law

(3) Isotropic spin (isospin, I)

p and n ~ I_3 = 1/2, –1/2.

π^+, π^0, and π^- ~ I_3 = 1, 0, –1

In different processes, ΔI and ΔI_3 obey different selection rules

For non-strange particles

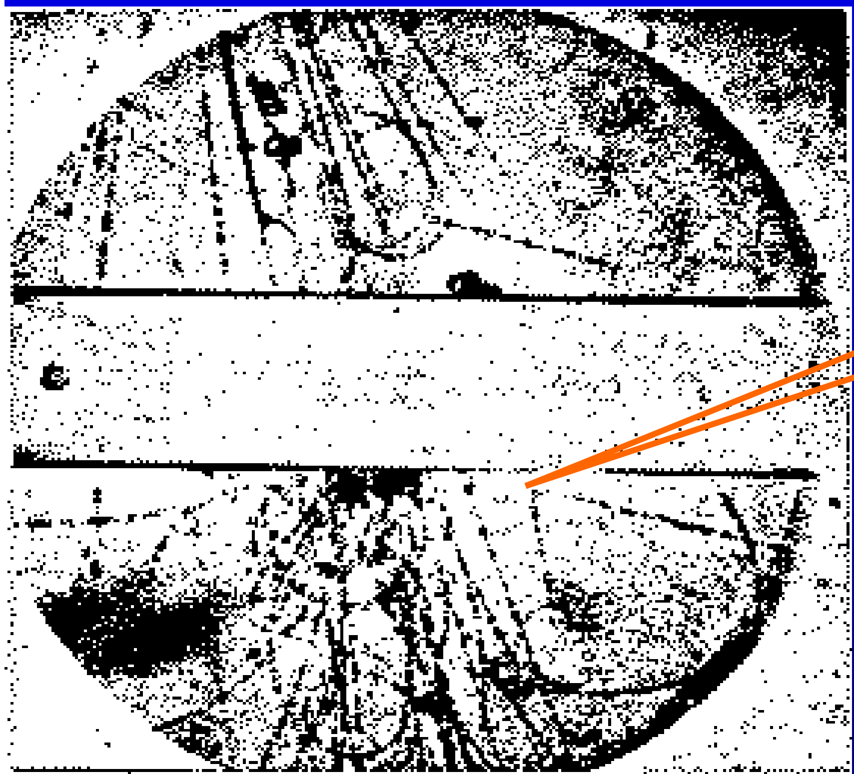
$$Q = I_3 + \frac{1}{2} B$$

The total isospin I is conserved in strong interactions

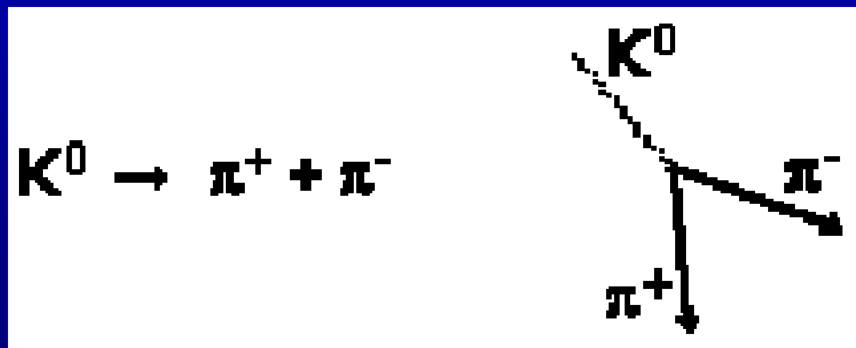
The total z-component I_3 is conserved in both strong and electromagnetic interaction.

4. Conservation of strangeness

In 1947, G. D. Rochester and C. C. Butler in Manchester
Cloud chamber



V particles



Particles of strong-production, weak-decay anomaly are called strange.

In 1947, $K^0 \to \pi^+ + \pi^-$ half-life 10^{-8}s

$\rho^0 \to \pi^+ + \pi^-$ half-life 10^{-23}s

This strangeness was explained by Gell-Mann and Pais, and independently Nisijima(西岛).

New quantum number strangeness S

Strange particles and their strangeness S are :

$$
\begin{array}{ll}
K^+, K^0: & 1 \\
K^-, \overline{K}^0: & -1 \\
\Lambda^0: & -1 \\
\Sigma^\pm, \Sigma^0: & -1 \\
\Xi^0, \Xi^-: & -2 \\
\Omega: & -3
\end{array}
$$

produced by strong interaction

$$\pi^- + p \rightarrow \Lambda^0 + K^0$$

$$0 \quad + \quad 0 = \quad -1 + 1 \qquad \text{strangeness}$$

$$p + p \rightarrow \Lambda^0 + K^+ + p$$

$$0 \quad + \quad 0 = \quad -1 + 1 \qquad \text{strangeness}$$

decayed by weak interaction

$$K^0 \rightarrow \pi^+ + \pi^-$$

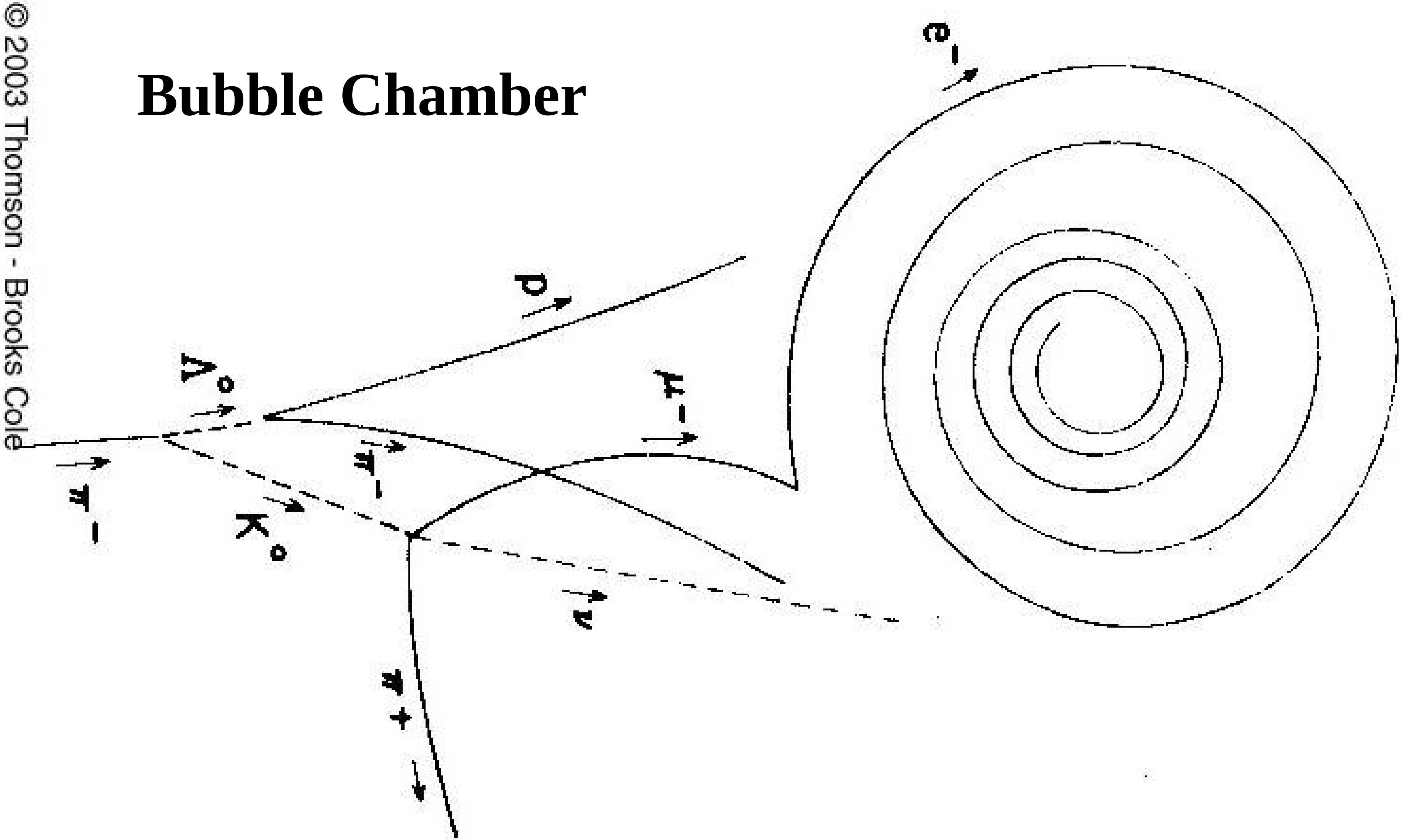
$$1 \neq 0 + 0 \qquad \text{strangeness}$$

$$\Lambda^0 \rightarrow \pi^- + p$$

$$-1 \neq 0 + 0 \qquad \text{strangeness}$$

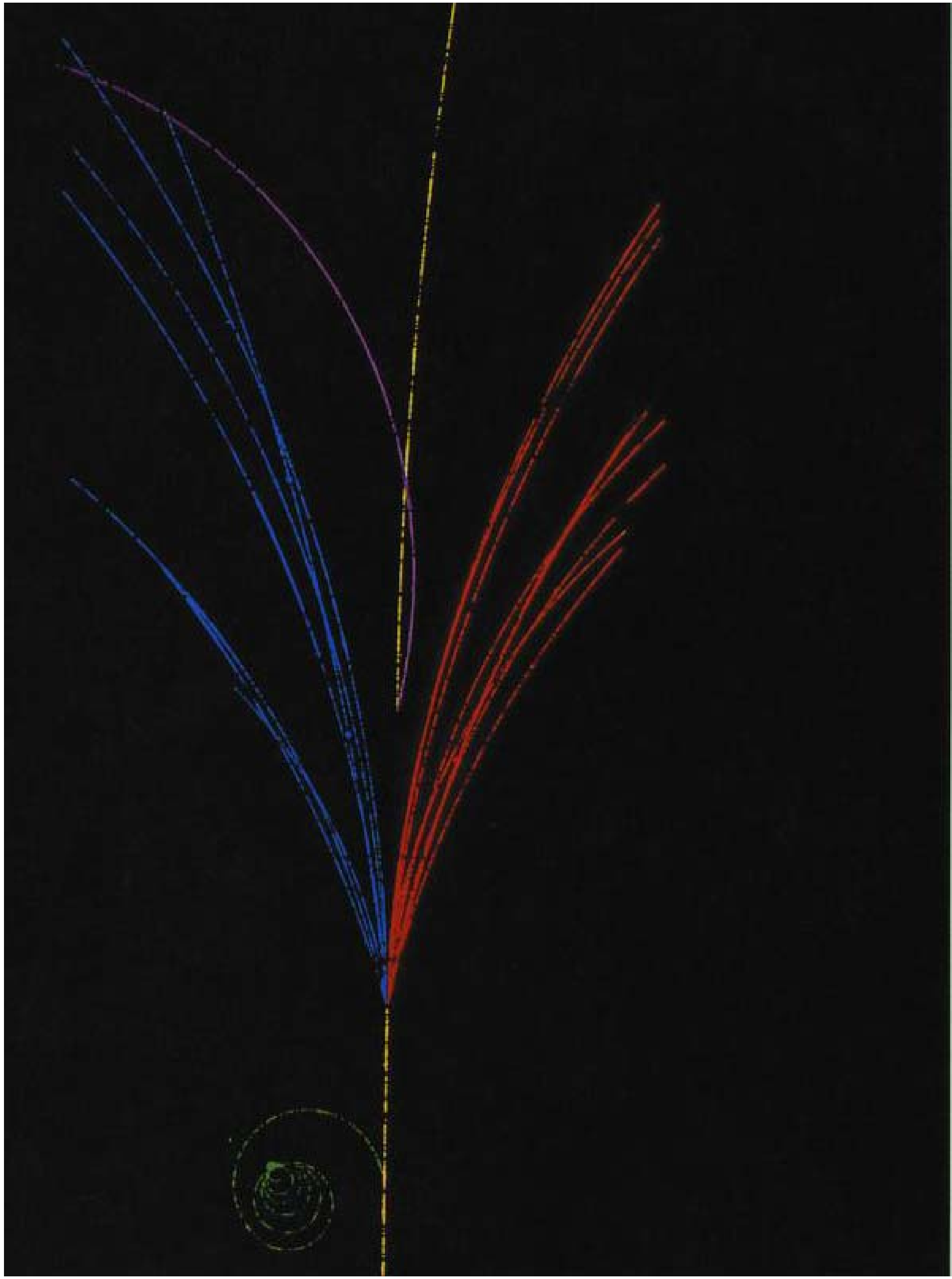
<u>The strange particles were produced quickly and in pairs but decayed slowly and decayed separately.</u>

Bubble Chamber



$\pi^- + p \rightarrow \Lambda_0 + K_0$ then $\Lambda_0 \rightarrow \pi^- + p$

and $K_0 \rightarrow \pi + \mu^- + \overline{\nu}_\mu$



liquid hydrogen bubble chamber at Brookhaven National Laboratory yellow line at the bottom is an incoming high-energy proton.

7 positive pions, a,proton, and a positive kaon to the right

seven negative pions to the left .

A neutral lambda is produced which travels upwards undetected and then
decays into a proton and a negative pion.

The total strangeness S is conserved in strong and electromagnetic interactions but not in weak interactions.

In process governed by the weak interaction, the total strangeness <u>either remain constant or changes by one unit.</u>

S is related to other quantum number via Gellmann-Nisijima formula

hypercharge

$$S = 2(Q - I_3) - B$$

$$Q = I_3 + \frac{B}{2} + \frac{S}{2} = I_3 + \frac{Y}{2}, \quad Y = B + S$$

Example problems:

1. An Ω^- is produced using a beam of K^- incident on protons. What other particles are produced in this reaction?

Solution:

$$K^- + p = \Omega^- + ?$$

$$\begin{array}{lcccccc} S & & -1 & = & -3 & + & 2 \\ B & 0 & +1 & = & 1 & + & 0 \\ Q & -1 & +1 & = & -1 & + & 1 \end{array}$$

Scanning through the tables of mesons and baryons, we find that we can satisfy the criteria with K^+ and K^0 . Therefore,

$$K^- + p = \Omega^- + K^+ + K^0.$$

2. How might Ω^- decay?

Solution: The Ω^- cannot decay by the strong interaction, because no final states with $S = -3$ are available. It must decay to particles have having total strangeness $S = -2$. Two possibilities are

$$\Omega^- \rightarrow \Lambda^0 + K^-,$$

and

$$\Omega^- \rightarrow \Xi^0 + \pi^-.$$

3. Determine if these processes are mediated by the strong or weak interaction, or are forbidden.

$$K^0 \rightarrow \pi^+ + \pi^-$$
$$1 \;\; = \; 0 \; + \;\; 0$$

weak interaction

$$K^- + p \rightarrow \Sigma^- + \pi^+$$
$$-1 + 0 = -1 + \;\; 0$$

strong interaction

$$K^- + p \rightarrow p + \pi^-$$
$$-1 + 0 \neq 0 + \;\; 0$$

forbidden

$$K^- + p \rightarrow \Lambda^0 + K^0$$
$$-1 + 0 \neq -1 + 1$$

forbidden

(5) There is no conservation law for meson.

Assignment: 30.4, 30.5 30.9, 30.10

30.4 The Quark Model

Robert Hofstadter

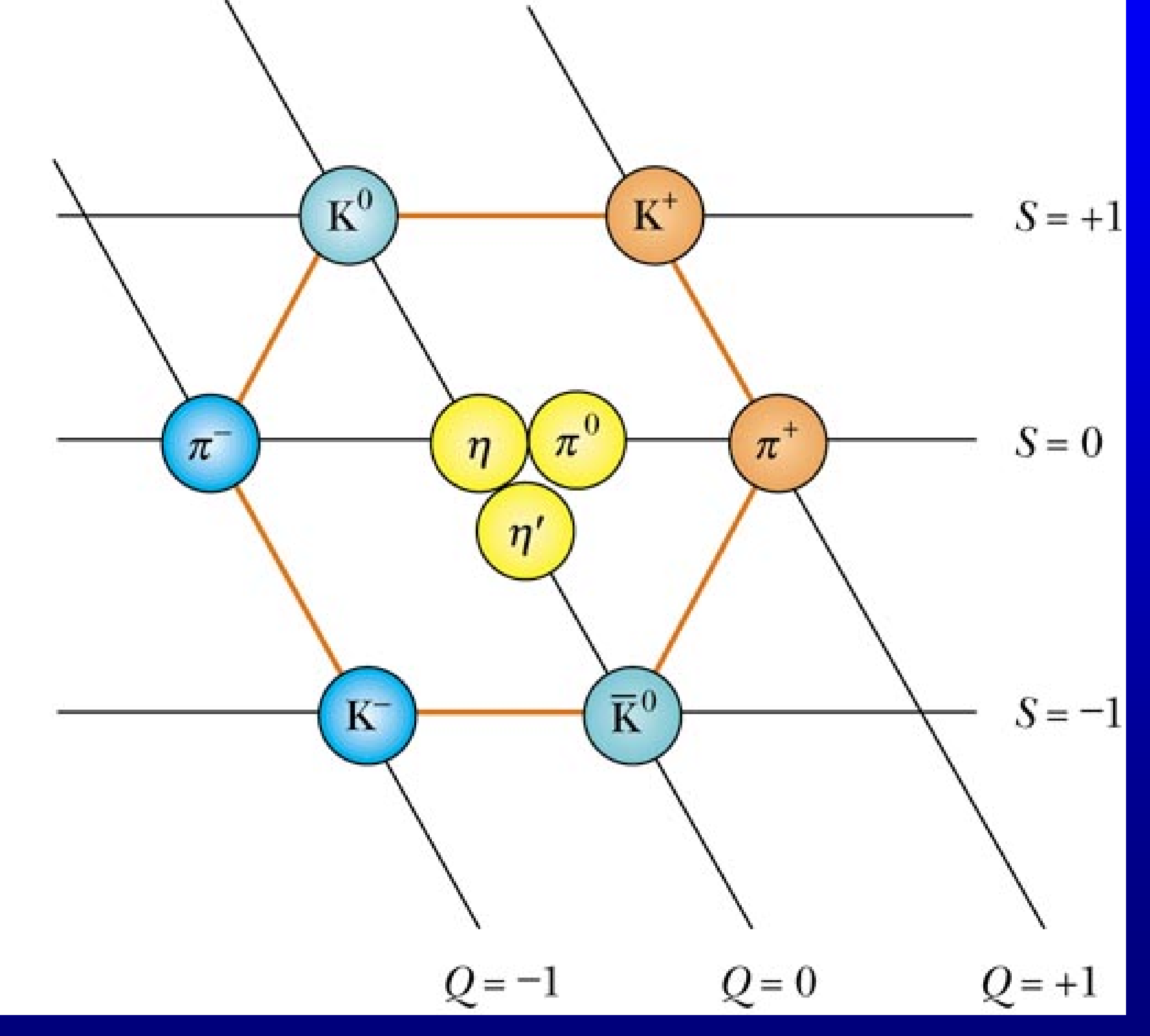
Structure of the proton and neutron

~0.8fm

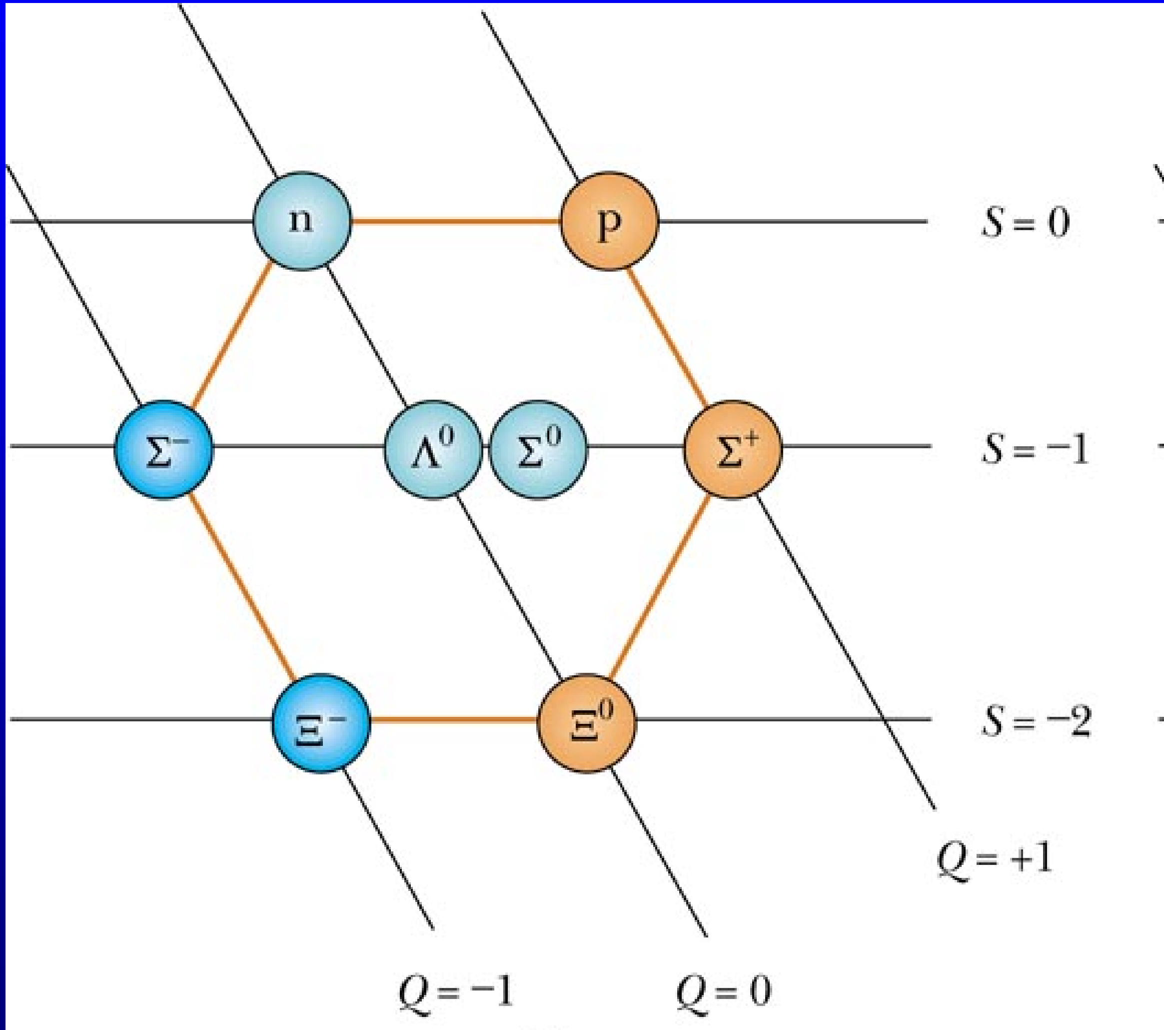
Murray Gell-mann and 8-fold way

10-fold way

An Eightfold Way for Mesons



An Eightfold Way for Baryons



“Three quarks for Muster Mark!/Sure he hasn't got much of a bark/And sure any he has it's all beside the mark.”

冲马克王呱叫三声!/很显然一声狗吠对他还不够/很显然他所有的一切都和盛名无关/

James Joyce‘s *Finnegans Wake* (芬尼根的彻夜祭）

Ulysses

George Zweig aces (幺点)

Basic assumptions of quark model

(1) Quarks are spin-1/2 particles.

(2)The baryon is composed of three quarks, written as qqq.

Since baryon number B is additive, we assign $B = 1/3$ to all quarks and –1/3 to antiquarks.

(3)The meson is composed of a quark and antiquark $q\bar{q}$. As its constituents can annihilate, this accounts for the fact that there are no absolutely stable meson states.

(4)Hadrons are characterized by the additive quantum number I_3, S, and C. We must assign appropriate values of I_3, S, and C to quarks.

At present there are six flavors of quarks:

- a pair of ordinary quarks "up"— u and "down" — d, that carry only isospin, but none of the other quantum numbers: u, $I_3 = 1/2$; d, $I_3 = -1/2$

- the strange quark s, which carries strangeness $S = -1$, but not charm or isospin

- the charmed quark c, which carries charm, but no strangeness or isospin, to which one assign $C = 1$ 粲

- the bottom or beauty quark b and the top or truth quark t. The fifth flavor is for $\Upsilon = b\bar{b}$.

Quark	Sym-bol	Spin	Charge	Bar-yon	S	C	B	T	Mass* (MeV)
Up	u	1/2	+2/3	1/3	0	0	0	0	3
Down	d	1/2	–1/3	1/3	0	0	0	0	6
Strange	s	1/2	–1/3	1/3	–1	0	0	0	100
Charm	c	1/2	+2/3	1/3	0	+1	0	0	1250
Top	t	1/2	+2/3	1/3	0	0	0	+1	175000
Bottom	b	1/2	–1/3	1/3	0	0	–1	0	4500

The fourth flavor c is to explain particle J/Ψ. New particle was first observed at Brookhaven(East coast) by a group under C. C. Ting group in the summer of 1974. But Ting wanted to check his results before announcing them publicly, and the discovery remained an astonishingly well-kept secret until the weekend of November 10-11, when the new particle was discovered independently by Burton Richter's group at SLAC(West coast). The two teams then published simultaneously. Ting named the particle J, and Richter called it Ψ. The mass of this particle is about 2~3 times of usual resonances, but its lifetime is 10^{-20}s, 10^3 times of the typical hadrons in this mass range. It might be a sign of fundamentally new physics. For good reason, the events precipitated by the discovery of this particle came to be known as the November Revolution.

1977 Leon Lederman discovers the Upsilon Υ

$\Upsilon \ (b\bar{b})$ bottom quark

In 1995, top quark was discovered at Fermilab Tevatron ; CDF and D0 detectors.

10^{12} eV

symbol	quark combination	mass MeV	spin	B	S
p	uud	938.3	1/2	1	0
n	ddu	939.6	1/2	1	0
Λ^0	uds	1115.6	1/2	1	-1
Σ^+	uus	1189.4	1/2	1	-1
Σ^0	uds	1192.5	1/2	1	-1
Σ^-	dds	1197.3	1/2	1	-1
Δ^{++}	uuu	1232	3/2	1	0

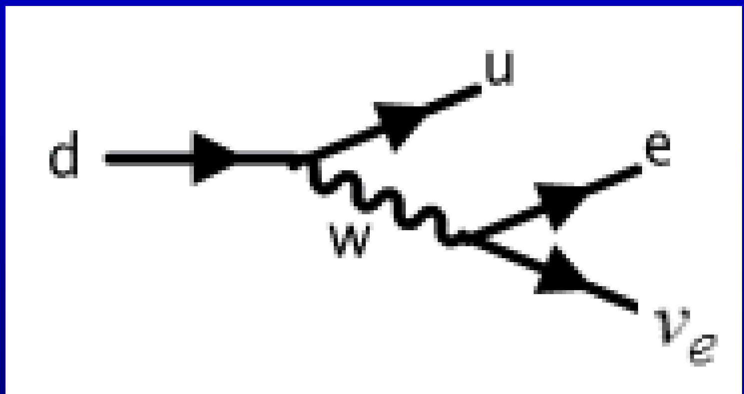
symbol	quark combination	mass MeV	spin	B	S
π^+	$u\bar{d}$	139.6	0	0	0
π^0	$(u\bar{u}+d\bar{d})/\sqrt{2}$	135.0	0	0	0
K^+	$u\bar{s}$	493.7	0	0	1

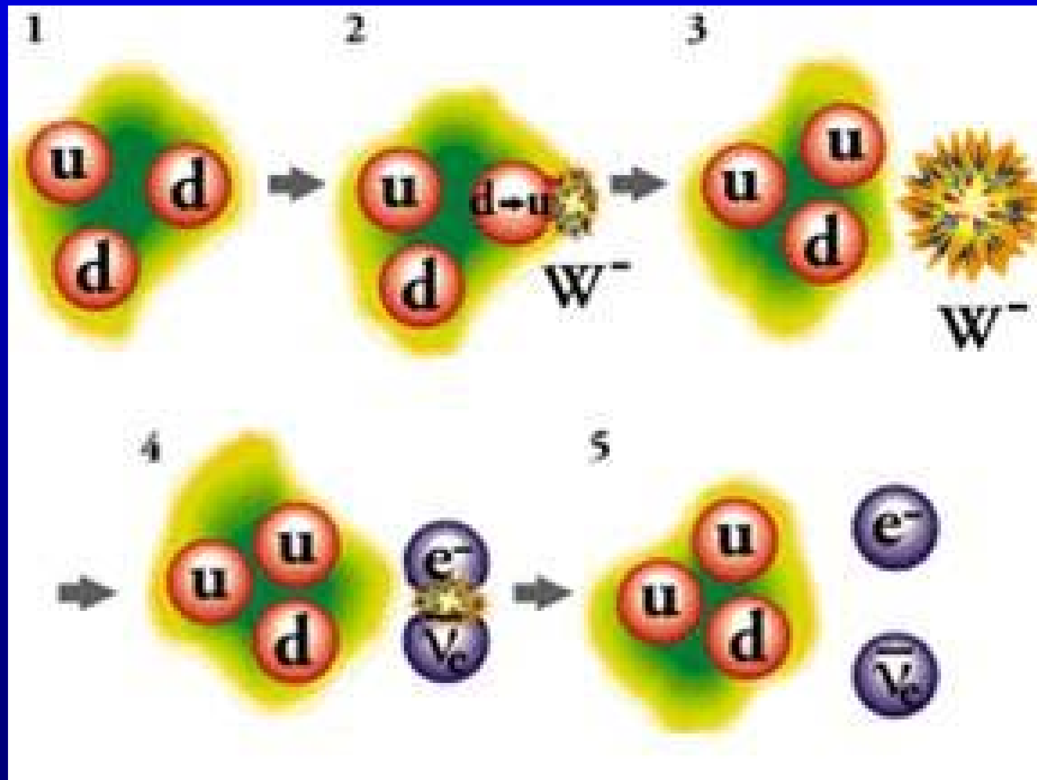
On the level of quarks, β decay can be understood as

$$udd \rightarrow uud + e^- + \bar{\nu}_e$$

$$d \rightarrow u + e^- + \bar{\nu}_e$$

$$d \rightarrow u + W^- \text{ and } W^- \rightarrow e^- + \bar{\nu}_e$$



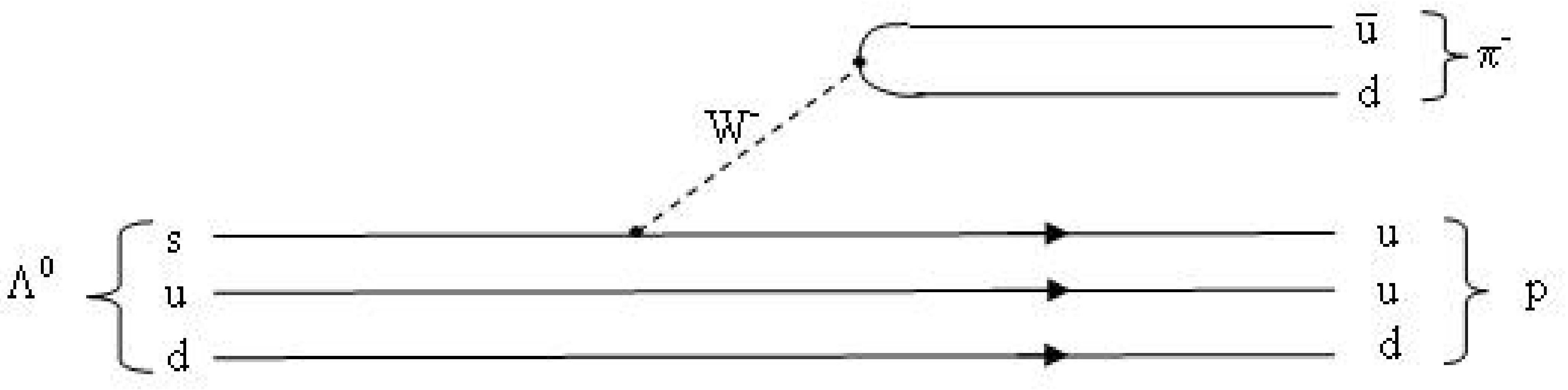


The process

$$\Lambda^0 \rightarrow p + \pi^-$$

can be viewed on the level of quarks as

$$sud \rightarrow uud + d\bar{u}.$$



The process

$$K^- + p \to \Omega^- + K^+ + K^0$$

can be written as

$$s\bar{u} + uud \to sss + u\bar{s} + d\bar{s}.$$

The net process is

$$u\bar{u} \to s\bar{s} + s\bar{s}.$$

The weak interaction can change one type of quark into another.

The strong interaction can create or destroy quark-antiquark pairs, but it cannot change one type of quark into another.

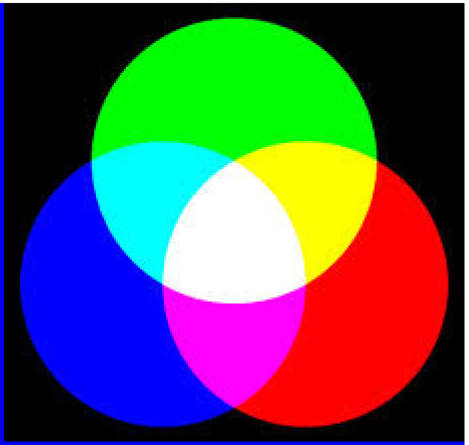
Color quantum number and quantum chromodynamics (QCD)

For Δ^{++}(uuu) and Ω^{-}(sss) the three quarks are identical and the wave function is totally symmetrical.
This is in contradiction to the spin-statistics theorem.

Greenberg (1964) suggested that there existed a hidden degree of freedom He called the proposed degree of freedom color. All naturally occurring particles are colorless or “white”.

Colored Quarks

- Color "charge" occurs in red, blue, or green
  - Antiquarks have colors of antired, antiblue, or antigreen
- Color obeys the Exclusion Principle
- A combination of quarks of each color produces white (or colorless)
- Baryons and mesons are always colorless (or white)

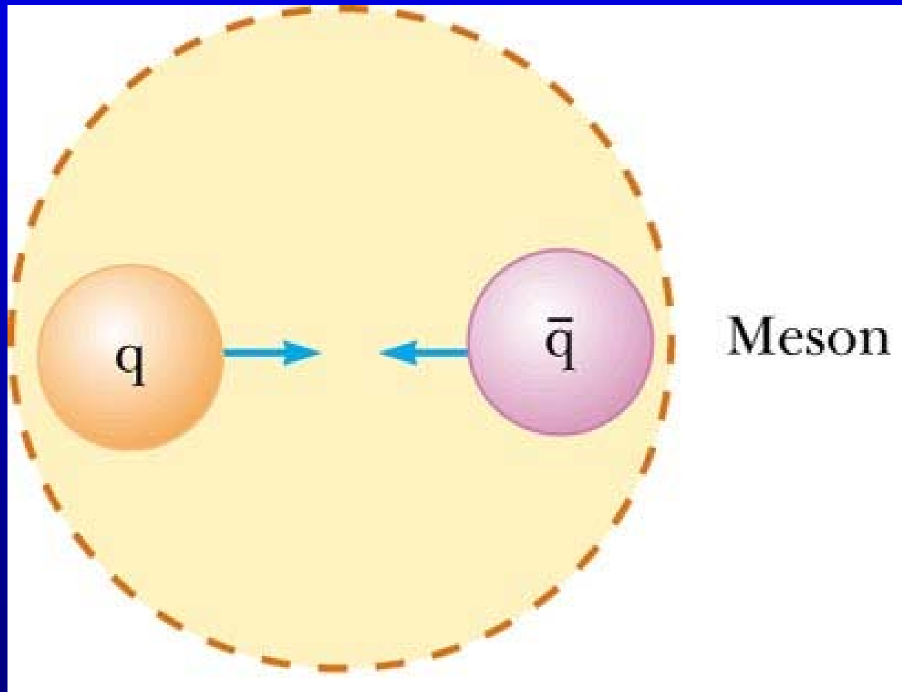


ternary neutral states **binary neutral states**

antired = white - red = cyan(青）
antigreen = white - green = magenta (洋红)
antiblue = white - blue = yellow

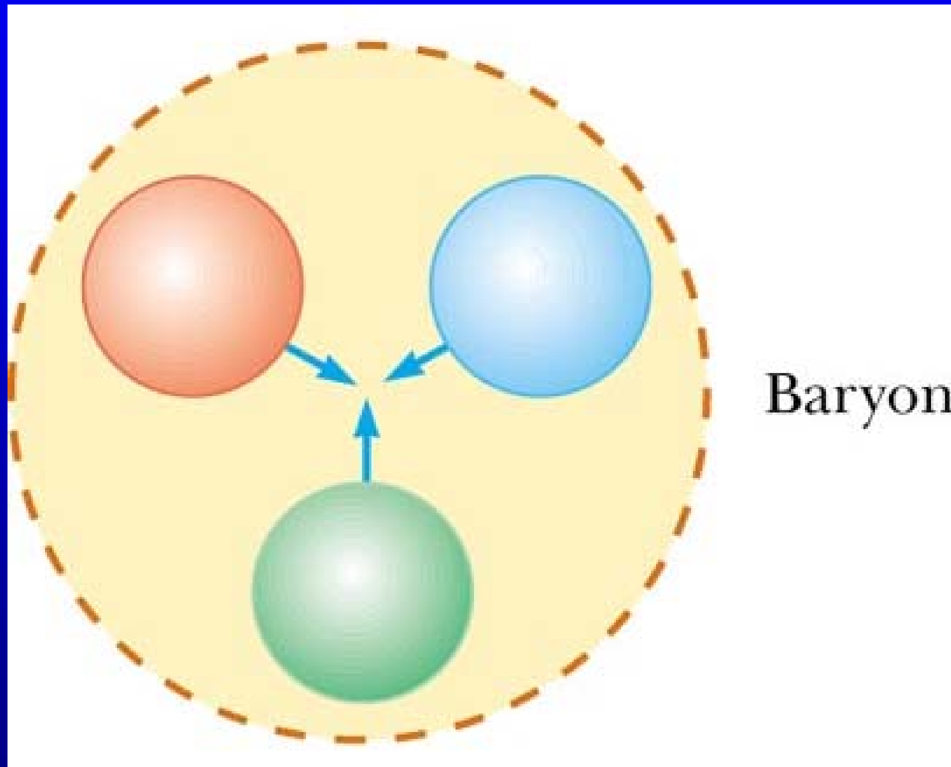
Mesons

- A red quark is attracted to an antired quark
- The quark – antiquark pair forms a meson
- The resulting meson is colorless



Baryon

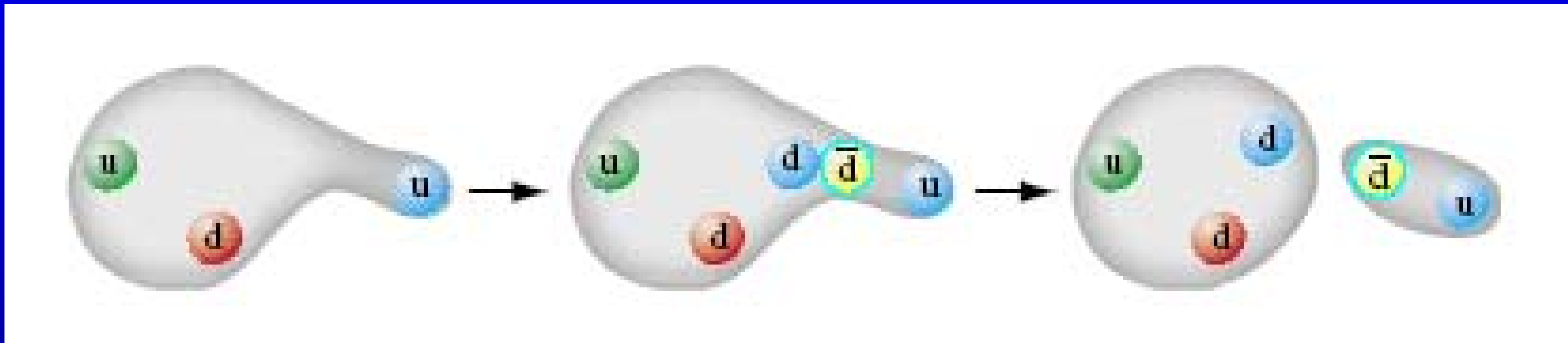
- Quarks of different colors attract each other
- The quark triplet forms a baryon
- The baryon is colorless



- QCD gave a new theory of how quarks interact with each other by means of color charge
- The strong force between quarks is often called the color force
- The strong force between quarks is carried by *gluons*
  - Gluons are massless particles
  - There are 8 gluons, all with color charge
- When a quark emits or absorbs a gluon, its color changes
- Like colors repel and unlike colors attract. The residual strong force is the nuclear force that binds the protons and neutrons to form nuclei

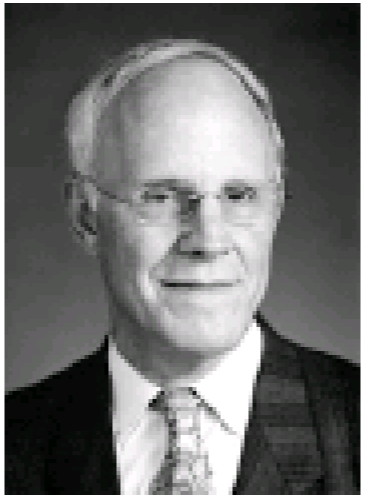
Asymptotic Freedom and Color Confinement

$$p \longrightarrow n + \pi^+$$



The Nobel Prize in Physics 2004

"for the discovery of asymptotic freedom in the theory of the strong interaction"



David J. Gross	H. David Politzer	Frank Wilczek
1/3 of the prize	1/3 of the prize	1/3 of the prize
USA	USA	USA
University of California, Kavli Institute for Theoretical Physics Santa Barbara, CA, USA	California Institute of Technology (Caltech) Pasadena, CA, USA	Massachusetts Institute of Technology (MIT) Cambridge, MA, USA
b. 1941	b. 1949	b. 1951

Standard Model of
FUNDAMENTAL PARTICLES AND INTERACTIONS

The Standard Model summarizes the current knowledge in Particle Physics. It is the quantum theory that includes the theory of strong interactions (quantum chromodynamics or QCD) and the unified theory of weak and electromagnetic interactions (electroweak). Gravity is included on this chart because it is one of the fundamental interactions even though not part of the "Standard Model."

FERMIONS matter constituents spin = 1/2, 3/2, 5/2, ...

Leptons spin = 1/2

Flavor	Mass GeV/c^2	Electric charge
ν_e electron neutrino	$<1\times10^{-8}$	0
e electron	0.000511	−1
ν_μ muon neutrino	<0.0002	0
μ muon	0.106	−1
ν_τ tau neutrino	<0.02	0
τ tau	1.7771	−1

Quarks spin = 1/2

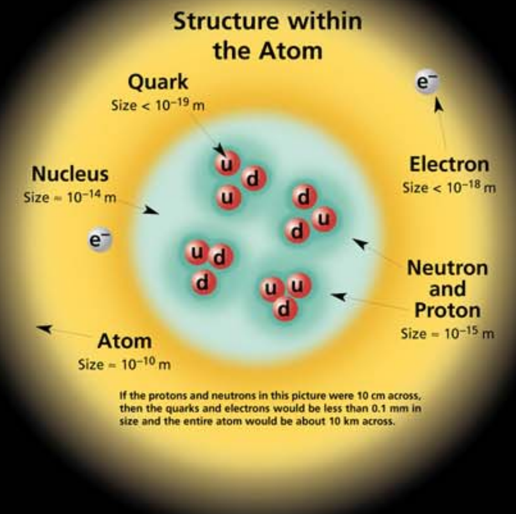
Flavor	Approx. Mass GeV/c^2	Electric charge
u up	0.003	2/3
d down	0.006	−1/3
c charm	1.3	2/3
s strange	0.1	−1/3
t top	175	2/3
b bottom	4.3	−1/3

Spin is the intrinsic angular momentum of particles. Spin is given in units of $\hbar$, which is the quantum unit of angular momentum, where $\hbar = h/2\pi = 6.58\times10^{-25}$ GeV s $= 1.05\times10^{-34}$ J s.

Electric charges are given in units of the proton's charge. In SI units the electric charge of the proton is 1.60×10^{-19} coulombs.

The **energy** unit of particle physics is the electronvolt (eV), the energy gained by one electron in crossing a potential difference of one volt. **Masses** are given in GeV/c^2 (remember $E = mc^2$), where 1 GeV $= 10^9$ eV $= 1.60\times10^{-10}$ joule. The mass of the proton is 0.938 GeV/c^2 $= 1.67\times10^{-27}$ kg.

Structure within the Atom



If the protons and neutrons in this picture were 10 cm across, then the quarks and electrons would be less than 0.1 mm in size and the entire atom would be about 10 km across.

BOSONS force carriers spin = 0, 1, 2, ...

Unified Electroweak spin = 1

Name	Mass GeV/c^2	Electric charge
γ photon	0	0
W^-	80.4	−1
W^+	80.4	+1
Z^0	91.187	0

Strong (color) spin = 1

Name	Mass GeV/c^2	Electric charge
g gluon	0	0

Color Charge

Each quark carries one of three types of "strong charge," also called "color charge." These charges have nothing to do with the colors of visible light. There are eight possible types of color charge for gluons. Just as electrically-charged particles interact by exchanging photons, in strong interactions color-charged particles interact by exchanging gluons. Leptons, photons, and ***W*** and ***Z*** bosons have no strong interactions and hence no color charge.

Quarks Confined in Mesons and Baryons

One cannot isolate quarks and gluons; they are confined in color-neutral particles called **hadrons**. This confinement (binding) results from multiple exchanges of gluons among the color-charged constituents. As color-charged particles (quarks and gluons) move apart, the energy in the color-force field between them increases. This energy eventually is converted into additional quark-antiquark pairs (see figure below). The quarks and antiquarks then combine into hadrons; these are the particles seen to emerge. Two types of hadrons have been observed in nature: **mesons** $q\bar{q}$ and **baryons** qqq.

Residual Strong Interaction

The strong binding of color-neutral protons and neutrons to form nuclei is due to residual strong interactions between their color-charged constituents. It is similar to the residual electrical interaction that binds electrically neutral atoms to form molecules. It can also be viewed as the exchange of mesons between the hadrons.

PROPERTIES OF THE INTERACTIONS

Property \ Interaction	Gravitational	Weak (Electroweak)	Electromagnetic (Electroweak)	Strong: Fundamental	Strong: Residual
Acts on:	Mass – Energy	Flavor	Electric Charge	Color Charge	See Residual Strong Interaction Note
Particles experiencing:	All	Quarks, Leptons	Electrically charged	Quarks, Gluons	Hadrons
Particles mediating:	Graviton (not yet observed)	W^+ W^- Z^0	γ	Gluons	Mesons
Strength relative to electromag for two u quarks at: 10^{-18} m	10^{-41}	0.8	1	25	Not applicable to quarks
Strength relative to electromag for two u quarks at: 3×10^{-17} m	10^{-41}	10^{-4}	1	60	Not applicable to quarks
for two protons in nucleus	10^{-36}	10^{-7}	1	Not applicable to hadrons	20

Baryons qqq and Antibaryons $\bar{q}\bar{q}\bar{q}$

Baryons are fermionic hadrons. There are about 120 types of baryons.

Symbol	Name	Quark content	Electric charge	Mass GeV/c^2	Spin
p	proton	uud	1	0.938	1/2
$\bar{p}$	anti-proton	$\bar{u}\bar{u}\bar{d}$	−1	0.938	1/2
n	neutron	udd	0	0.940	1/2
Λ	lambda	uds	0	1.116	1/2
Ω^-	omega	sss	−1	1.672	3/2

Mesons $q\bar{q}$

Mesons are bosonic hadrons. There are about 140 types of mesons.

Symbol	Name	Quark content	Electric charge	Mass GeV/c^2	Spin
π^+	pion	$u\bar{d}$	+1	0.140	0
K^-	kaon	$s\bar{u}$	−1	0.494	0
ρ^+	rho	$u\bar{d}$	+1	0.770	1
B^0	B-zero	$d\bar{b}$	0	5.279	0
η_c	eta-c	$c\bar{c}$	0	2 .980	0

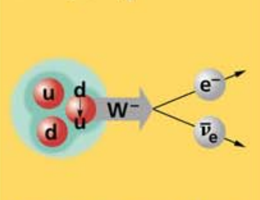
Matter and Antimatter

For every particle type there is a corresponding antiparticle type, denoted by a bar over the particle symbol (unless + or – charge is shown). Particle and antiparticle have identical mass and spin but opposite charges. Some electrically neutral bosons (e.g., Z^0, γ, and $\eta_c = c\bar{c}$, but not $K^0 = d\bar{s}$) are their own antiparticles.

Figures

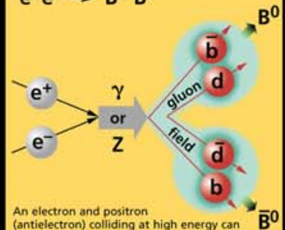
These diagrams are an artist's conception of physical processes. They are *not* exact and have *no* meaningful scale. Green shaded areas represent the cloud of gluons or the gluon field, and red lines the quark paths.

$n \rightarrow p\, e^-\, \bar{\nu}_e$



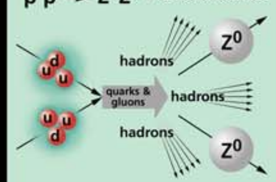
A neutron decays to a proton, an electron, and an antineutrino via a virtual (mediating) W boson. This is neutron β decay.

$e^+e^- \rightarrow B^0\, \bar{B}^0$



An electron and positron (antielectron) colliding at high energy can annihilate to produce B^0 and $\bar{B}^0$ mesons via a virtual Z boson or a virtual photon.

$p\, p \rightarrow Z^0 Z^0$ + assorted hadrons



Two protons colliding at high energy can produce various hadrons plus very high mass particles such as Z bosons. Events such as this one are rare but can yield vital clues to the structure of matter.

The Particle Adventure

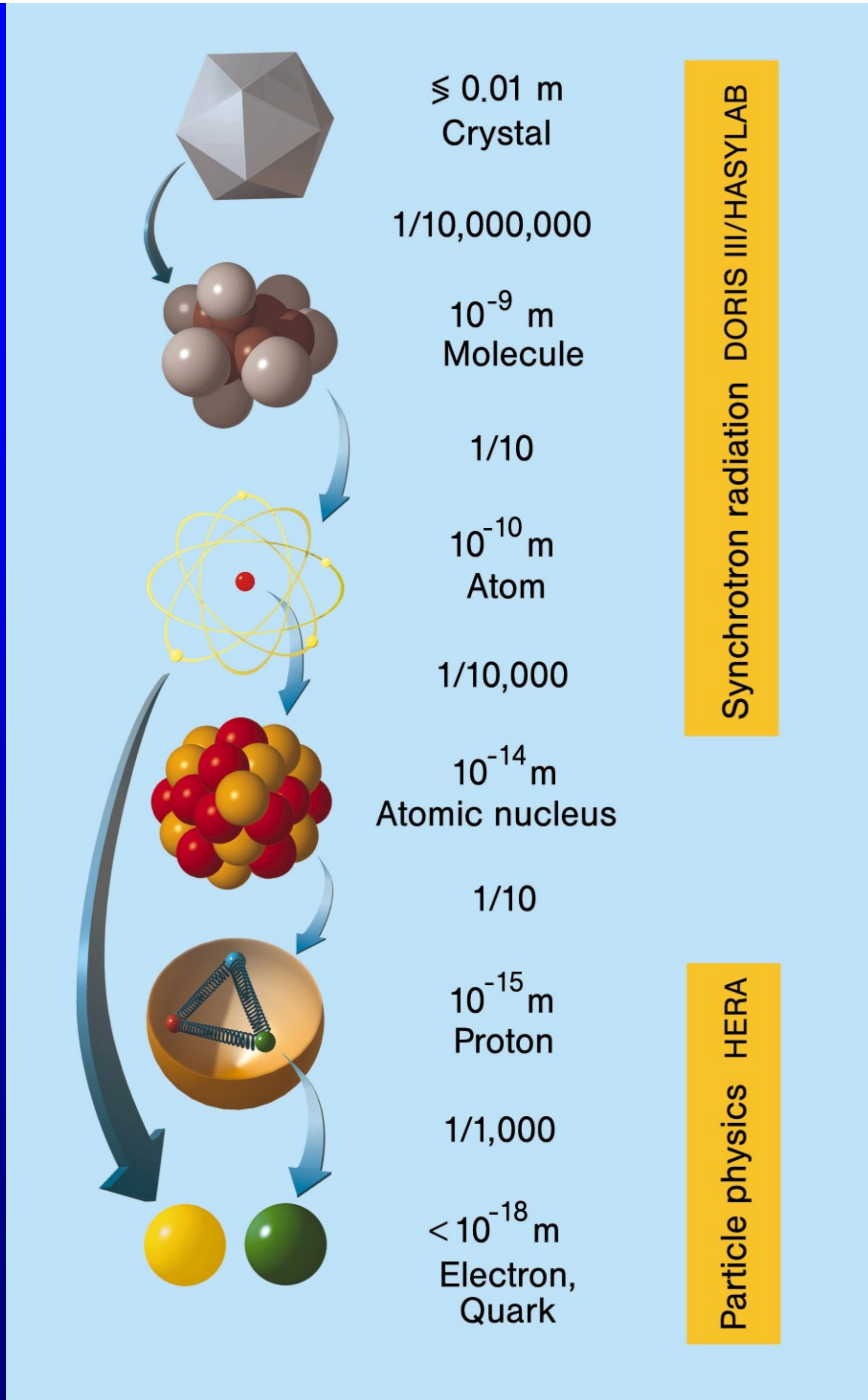
Visit the award-winning web feature *The Particle Adventure* at **http://ParticleAdventure.org**

This chart has been made possible by the generous support of:
U.S. Department of Energy
U.S. National Science Foundation
Lawrence Berkeley National Laboratory
Stanford Linear Accelerator Center
American Physical Society, Division of Particles and Fields
BURLE INDUSTRIES, INC.

http://CPEPweb.org

virus, 100 nm
atom, 0.1nm 100pm
nucleus, 10 fm
proton, 1 fm
quark, <1 am

nanometer 10^{-9}
picometer 10^{-12}
femtometer 10^{-15}
attometer 10^{-18}
zeptometer 10^{-21}
yoctometer 10^{-24}



30.5 CPT Theorem

CPT theorem was proposed by Schwinger (1953), Luders (1954), and Pauli (1955). It reads

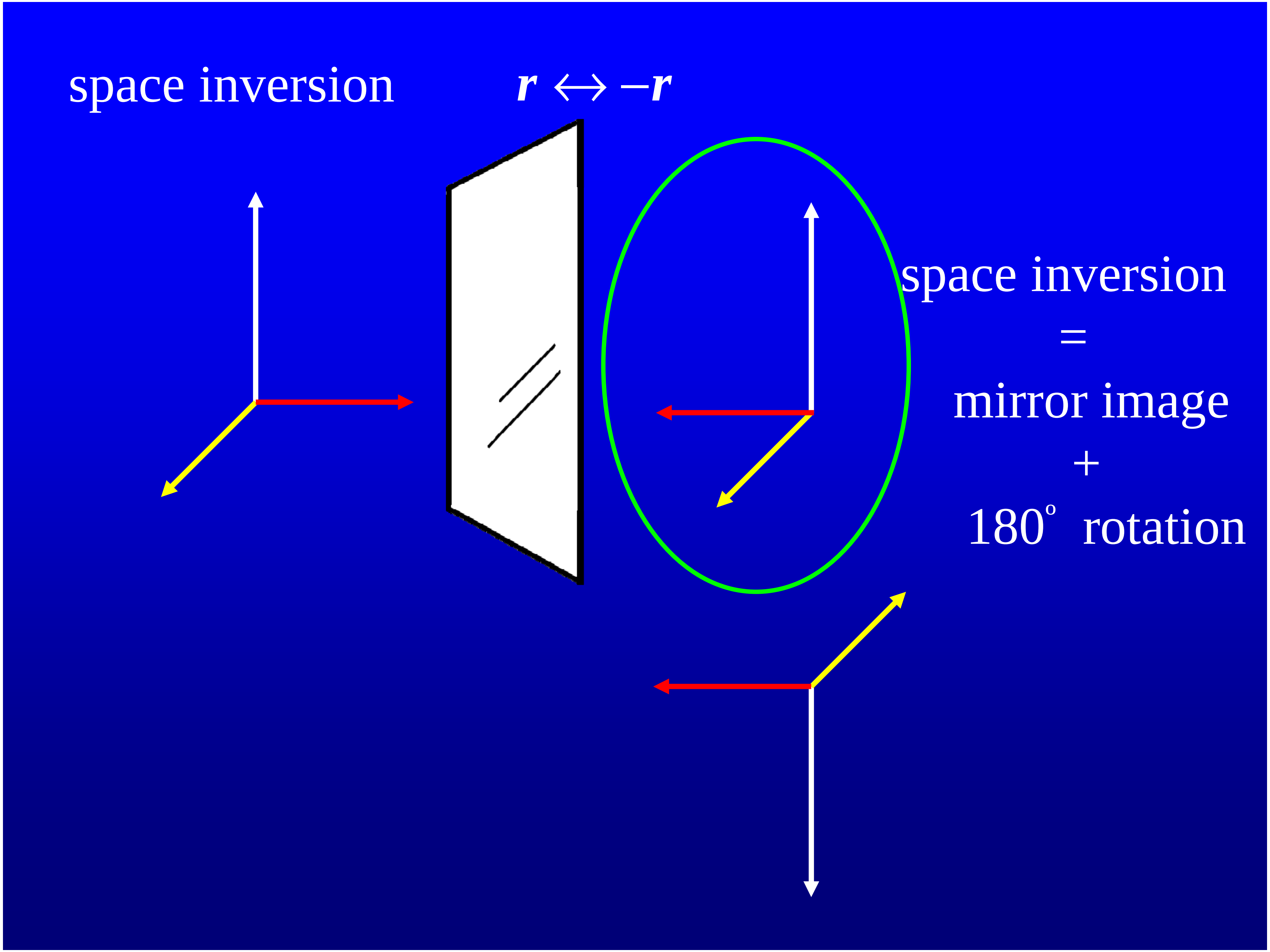
- Physical laws are invariant under combined operation of CPT.

C — charge conjugate operation.

particle — antiparticle

P — space inversion invariance. (<u>a mirror image plus a rotation</u>). parity operation .

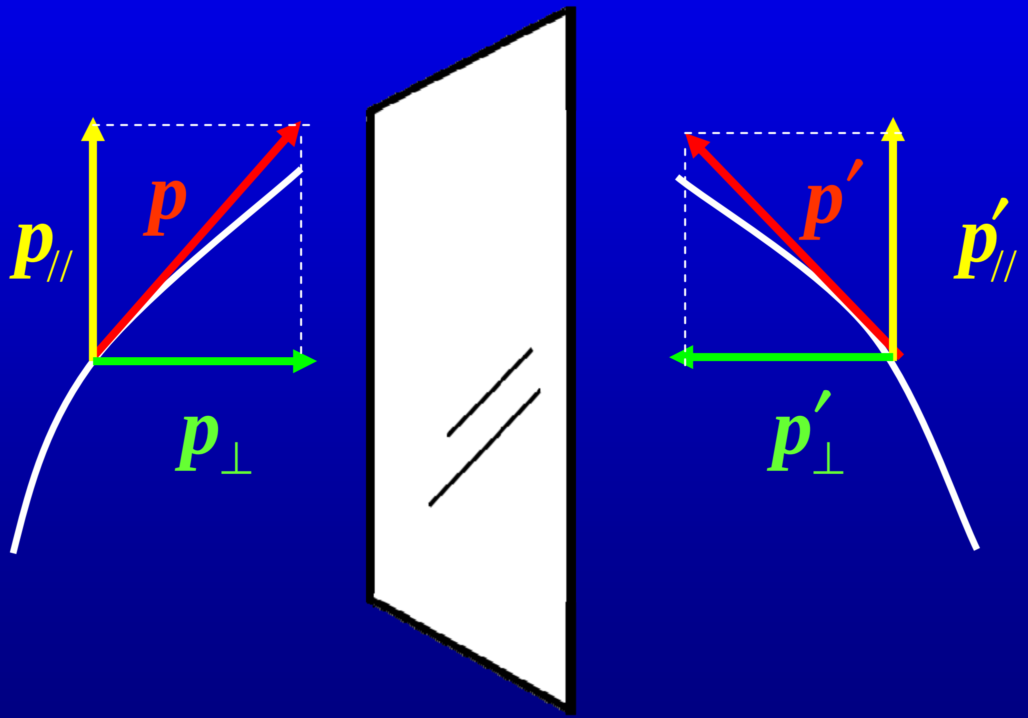
T — time reversal operation.


space inversion
$\boldsymbol{r} \leftrightarrow -\boldsymbol{r}$
space inversion
=
mirror image
+
180° rotation

polar vectors

axial vectors (pseudo vectors)

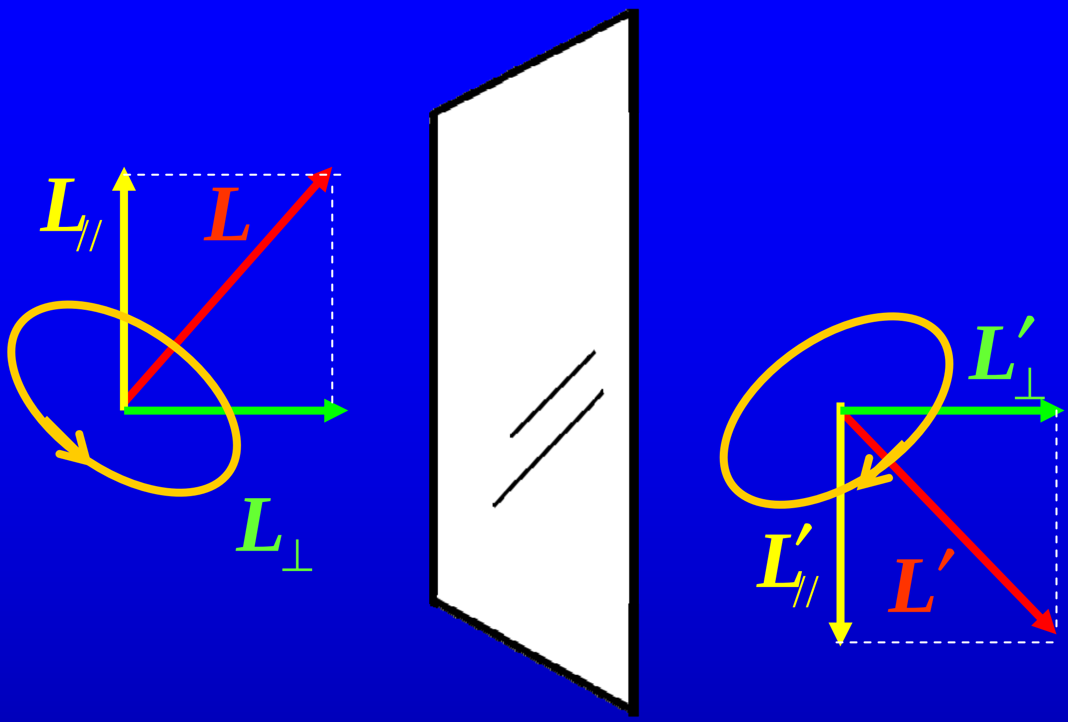
momentum



polar vector

$$p_{\perp} = -p'_{\perp}$$

$$p_{//} = p'_{//}$$



axial vector
(pseudo vector)

$$L_{\perp} = L'_{\perp}$$

$$L_{//} = -L'_{//}$$

Polar vectors: displacement, velocity, electric field, electric current density , or momentum

Axial vectors: angular velocity, magnetic field, magnetization, or angular momentum

In electrical-mechanical problem, C, P, and T conserve respectively.

$$\frac{d\boldsymbol{p}}{dt} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$$

parity: $\boldsymbol{r} \leftrightarrow -\boldsymbol{r}$

$$\boldsymbol{p} \leftrightarrow -\boldsymbol{p} \quad \boldsymbol{E} \leftrightarrow -\boldsymbol{E} \quad \boldsymbol{v} \leftrightarrow -\boldsymbol{v} \quad \boldsymbol{B} \leftrightarrow \boldsymbol{B}$$

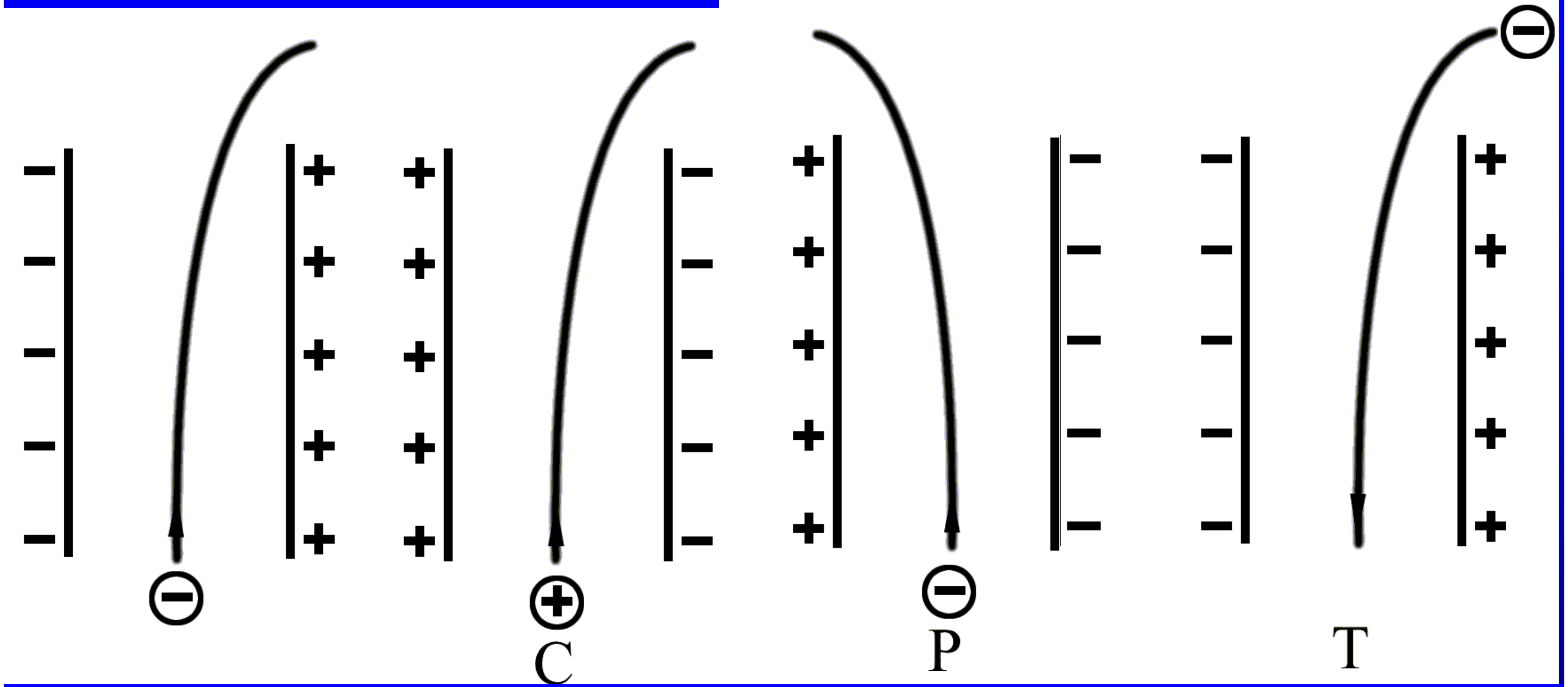
$$\frac{d(-\boldsymbol{p})}{dt} = -q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$$

charge conjugate: $q \leftrightarrow -q \quad \boldsymbol{E} \leftrightarrow -\boldsymbol{E} \quad \boldsymbol{B} \leftrightarrow -\boldsymbol{B}$

time reversal: $t \leftrightarrow -t$

$$t \leftrightarrow -t \quad \boldsymbol{p} \leftrightarrow -\boldsymbol{p} \quad \boldsymbol{E} \leftrightarrow \boldsymbol{E} \quad \boldsymbol{v} \leftrightarrow -\boldsymbol{v} \quad \boldsymbol{B} \leftrightarrow -\boldsymbol{B}$$

$$\frac{d(-\boldsymbol{p})}{d(-t)} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$$


C
P
T

Parity conservation

Parity is conserved in a process if the mirror image of the process is also a process that can occur in nature.

Before 1956, people believed C, P, and T conserve respectively. In thermodynamics, some phenomena against T are due to statistical law. But we are talking about fundamental particles!

it started from θ – τ puzzle

$$\tau^+ \rightarrow \pi^+\pi^+\pi^-, \pi^+\pi^0\pi^0$$

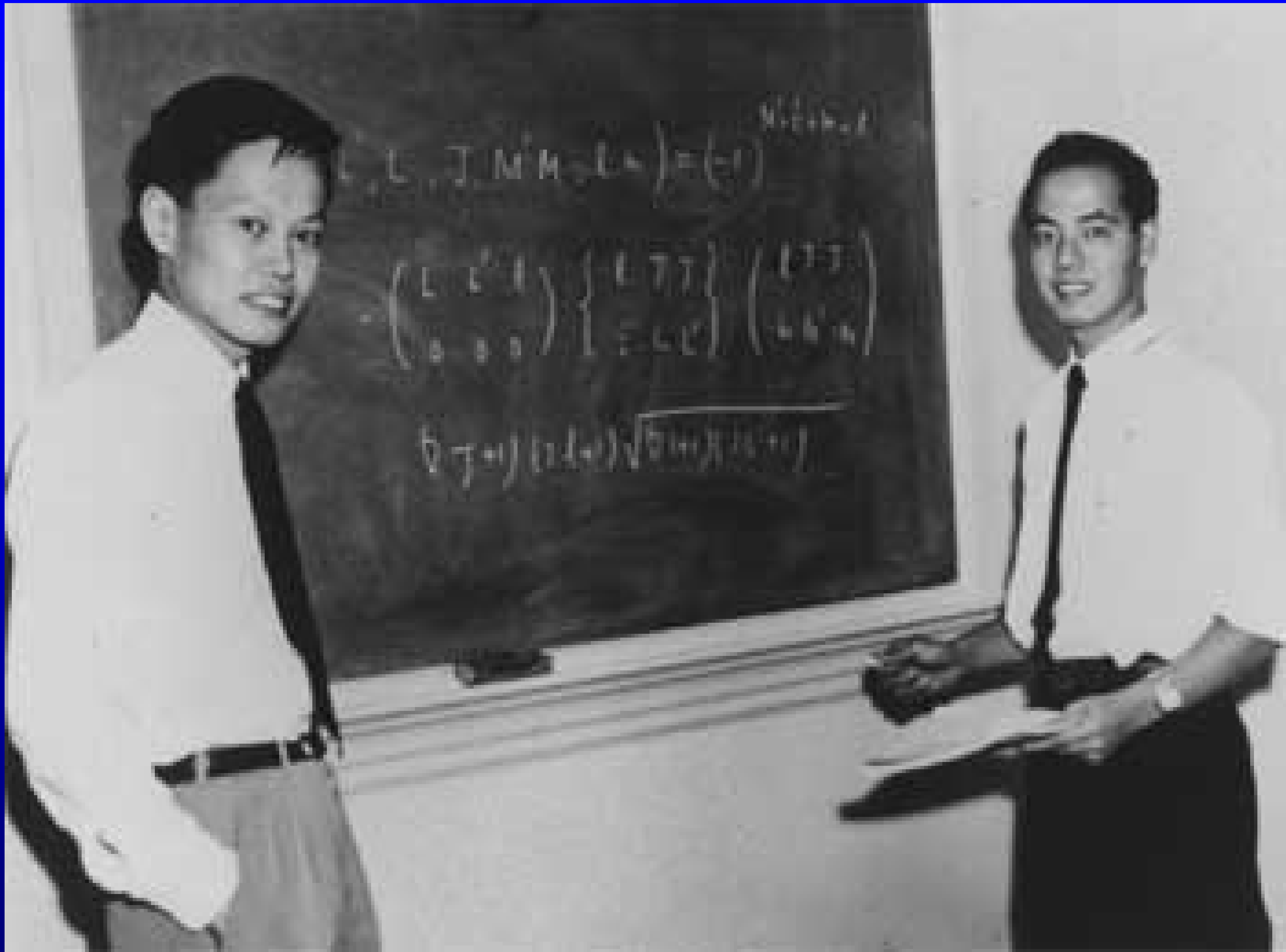
$$\theta^+ \rightarrow \pi^+\pi^0$$

Pion has a negative intrinsic parity. Quarks are taken to have positive intrinsic parity, antiquarks negative. Parity is multiplicative instead of "additive". The combination $q\bar{q}$ then has a negative parity.

$$\tau^+ , \theta^+ \text{-------} K^+$$

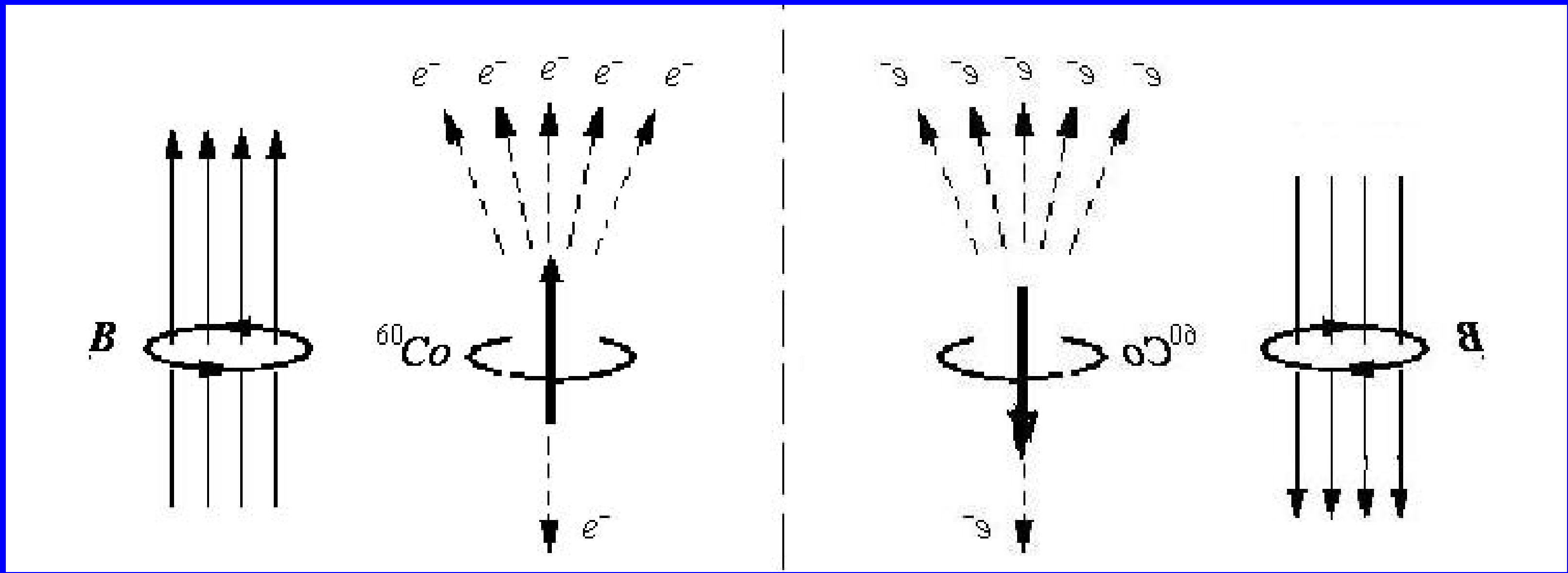
The state 2π has positive parity and 3π negative. If parity conserves in the process, θ and τ must be different particles. But they have same mass, spin, charge, lifetime and so on. If they are the same particle, the (weak) decay processes do not obey parity conservation law.

1956 T. D. Lee and C. N. Yang thoroughly investigated all weak interaction experiments ever done till that time, and found that there was no one that has proved parity conservation. They proposed in weak interaction parity could be non-conserved.



Wu et al. (Chien-Shiung Wu, Ambler, Hayward, Hoppes and Hudson) showed asymmetry in the β-ray counting rate in the experiments

A magnetic field is used to align magnetic moments (thence spins) of ^{60}Co. $T \sim 0.01$K to reduce the thermal agitation.



original process mirror image process

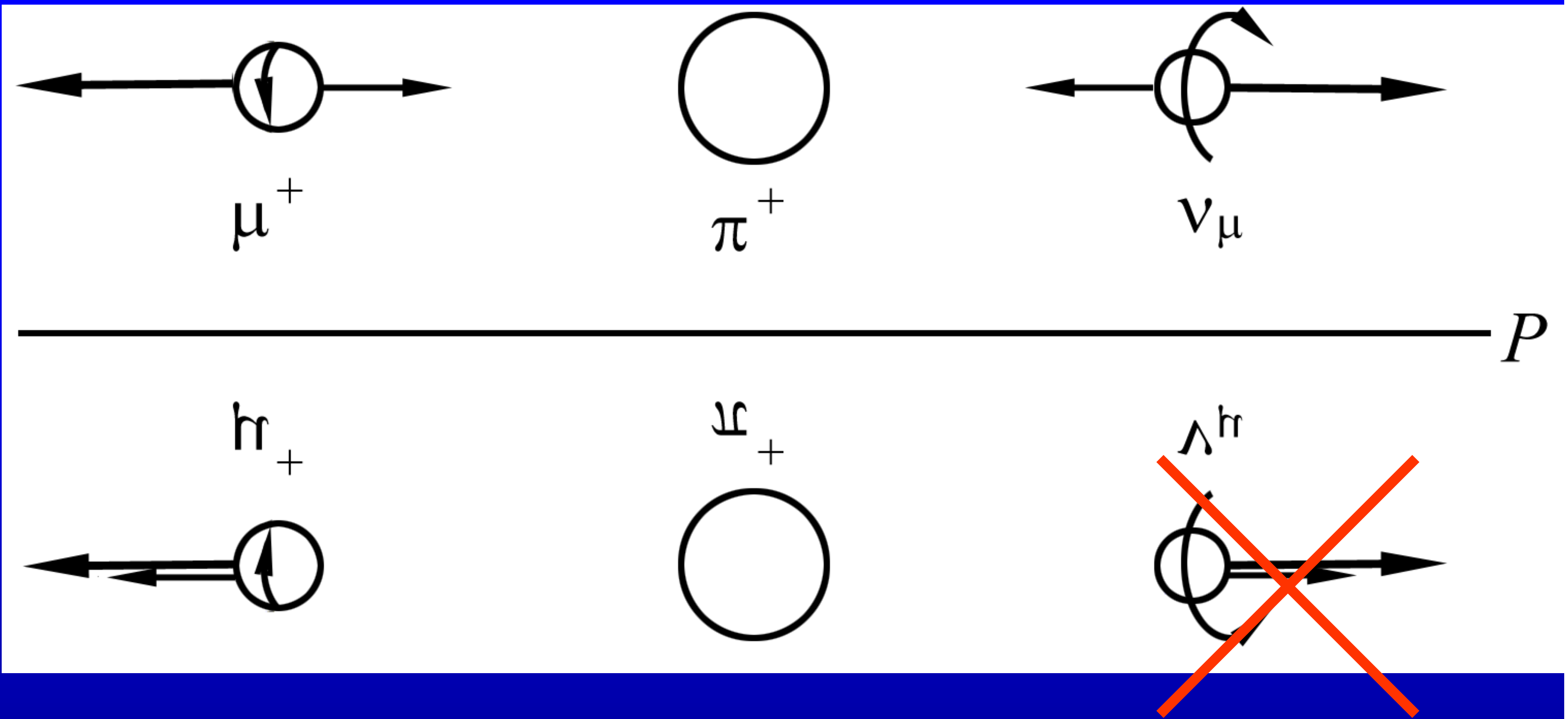
The mirror image process cannot occur in nature. Therefore, the parity is not conserved in the process.

Is it possible such a symmetry breaking is due to some our ignorance about the “essence”. We may try CP. The difficulty is no ^{60}Co anti-nucleus available

Charge Conjugate

A system is invariant with regard to the charge conjugation when, if a process is possible in the original system is also in conjugate system.

$$\pi^+ \rightarrow \mu^+ + \nu_\mu$$

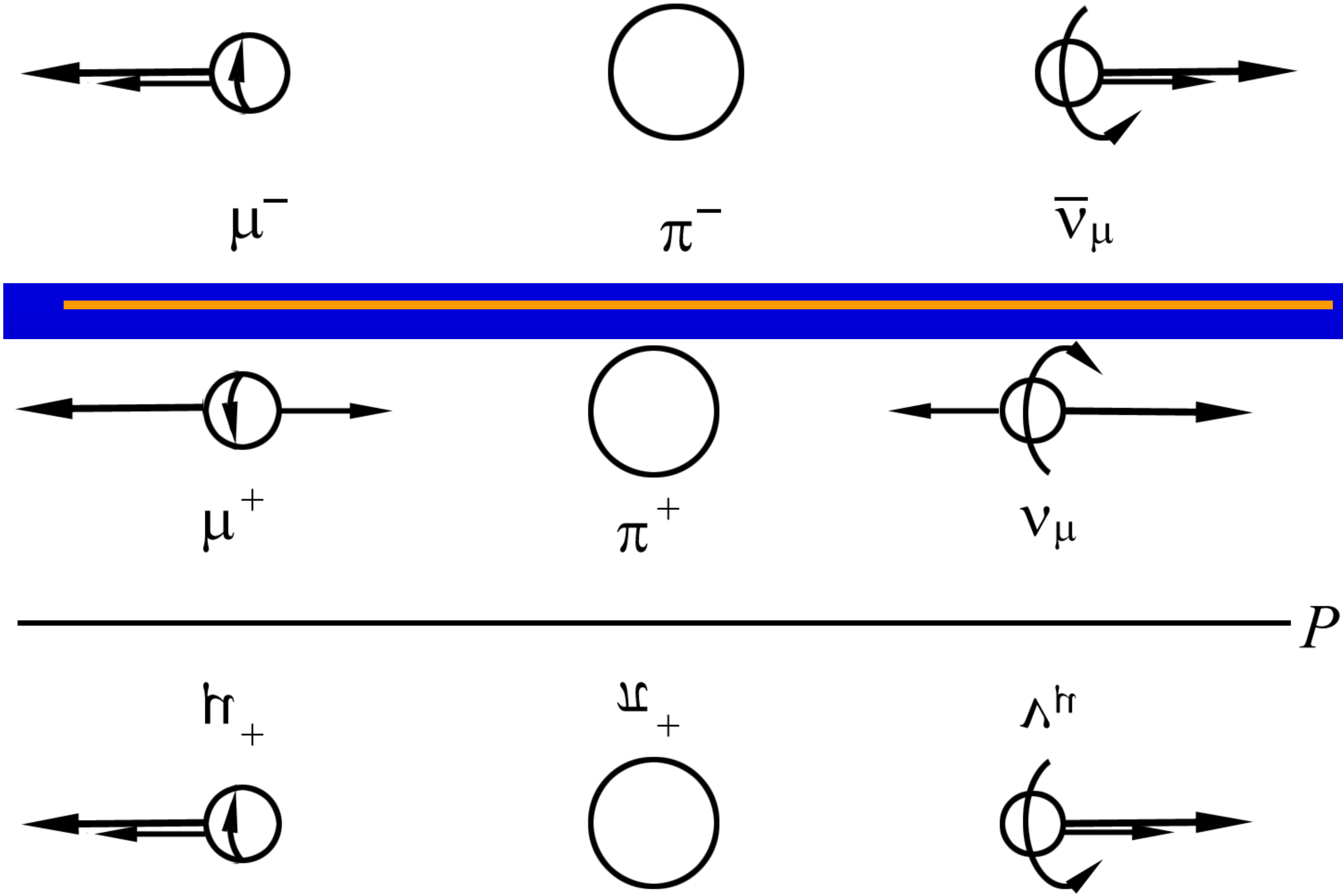


helicity: sgn($p \cdot s$)

handness

The helicity for a neutrino is –1 and the helicity for an antineutrino is +1.!!!

CP yields a possible process:



CP is slightly non-conserved.

In 1965, a group working at Brookhaven synchrotron found a small violation of CP invariance by carefully measuring decay modes in the decay of K^0.

$$(K^0 \text{ or } \overline{K}^0) \rightarrow \begin{matrix} e^+ + \nu + \pi^- \\ e^- + \overline{\nu} + \pi^+ \end{matrix}$$

It turns out that the e^+ mode is slightly preferred over e^-.

$$\delta = \frac{\Gamma(\mathrm{e}^+) - \Gamma(\mathrm{e}^-)}{\Gamma(\mathrm{e}^+) + \Gamma(\mathrm{e}^-)} = (3.30 \pm 0.12) \times 10^{-3}$$

People expect that the violation of CP conservation is the reason why matter is more than anti-matter.

At last we can define “positive” charge by the preferred mode of K^0

International Conference on High Energy Physics at the University of Rochester in April 1956.

Lee and Yang attended the conference with a proposal for ending the theta-tau puzzle. Their idea was that certain kinds of elementary particles occur in two forms with different parities. The idea was called *parity doubling*.

Also attending the conference was the theoretical physicist Richard Feynman. Feynman's roommate at the conference was the experimentalist Martin Block. Block suggested to Feynman on the first night of the conference that parity just may not be conserved in certain interactions.

The next day, following Yang's presentation of the parity doubling idea, Feynman brought up the question of non-conservation of parity. Feynman himself later said, "I thought the idea (of parity violation) unlikely, but possible, and a very exciting possibility." Indeed Feynman later made a fifty dollar bet with a friend that parity would not be violated . Yang's reply was that he and Lee had considered the idea but had arrived at no conclusions. During the discussion, Wigner, who had formulated the law of conservation of parity in the first place, also suggested that perhaps it did not hold in weak interactions .

Lee and Yang pursued the question further after the conference. "Early in May, when they were sitting in the White Rose Cafe near the corner of Broadway and 125th Street, in the vicinity of Columbia University, it suddenly struck them that it might be profitable to make a careful study of all known experiments involving weak interactions" . After several weeks of reviewing past experiments, they had come to two conclusions:

"Past experiments on the weak interactions had actually no bearing on the question of parity conservation."

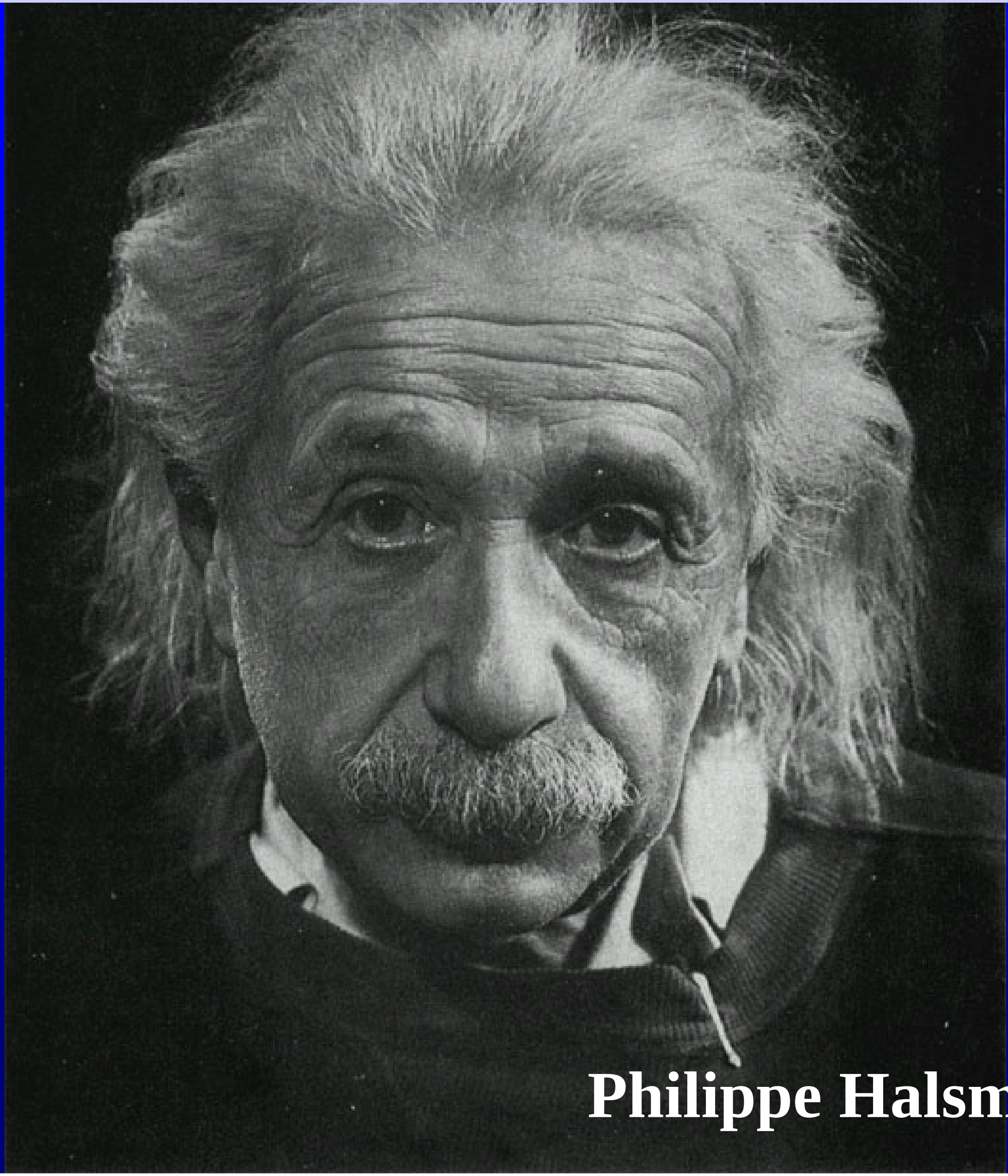
"In strong interactions, ... there were indeed many experiments that established parity conservation to a high degree of accuracy..."

Even before Lee and Yang's paper had been submitted to *The Physical Review*, Lee had discussed the experiment with Wu. At the time, Wu and her husband had planned a trip to Europe and the Far East. But she chose instead to remain and perform the experiment rather than lose the opportunity to other physicists who might recognize its importance.

Wu et al. (Chien-Shiung Wu, Ambler, Hayward, Hoppes and Hudson) showed asymmetry in the β-ray counting rate in the experiments

A magnetic field is used to align magnetic moments (thence spins) of ^{60}Co. $T \sim 0.01$K to reduce the thermal agitation.

PERSON OF THE CENTURY



Philippe Halsman (1948)

The most incomprehensible thing about the world is that it is comprehensible.

The most beautiful experience we can have is the mysterious. It is the fundamental emotion that stands at the cradle of true art and true science. Whoever does not know it and can no longer wonder, no longer marvel, is as good as dead, and his eyes are dimmed.

Albert Einstein

Thank you

for your attentions and efforts !